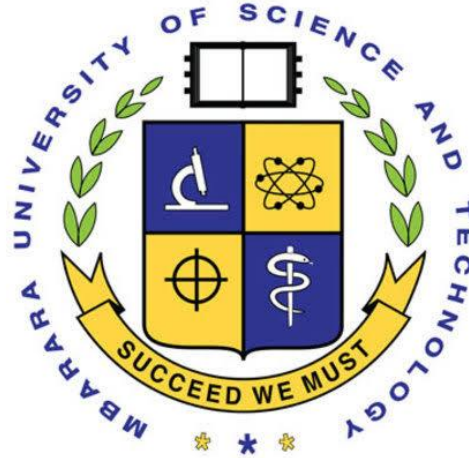


MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY



FACULTY OF COMPUTING AND INFORMATICS

Department of Computer Science

The Revised Bachelor of Science Computer Science (BSc. CS) Degree Programme

Day (Full Time) Program

October, 2019

Summary

The Mbarara University of Science and Technology (MUST) BSc. CS curriculum is designed to provide an effective integration of practical skills and industrial experience well backed by theoretical knowledge in key aspects of computing and technology. This rich curriculum includes a Mandatory 8-week industrial placement. Speciality training will be carried out during the recess terms of first year. There are three speciality training courses for professional certifications that is 1) networks and systems 2) database management systems, and 3) software development.

This curriculum provides strong foundation for producing graduates with sound knowledge and skills in computing and technology with ability to plan, design, implement and manage a wide range of digital computing systems in the computing world. Besides, this curriculum provides two areas for emphasis namely, Systems/Networks Engineering and Software Engineering/Development. Students who choose Networks emphasis take mandatory courses of Computer Networks, Data Communication, Telecommunications & Computing, Logic Design, Mobile Computing as well as other Computer Science elective courses. Students who choose Software Development emphasis must take mandatory courses of Programming Methods, Object Oriented Programming, Data Structures and Algorithms, Compiler Construction, Software Engineering, Mobile Applications Development, Artificial Intelligence and Data Science. These are aimed at providing a wider choice for specialisation in career and further studies.

The programming courses offered in year one and two, help students to gain skills in software development and build special interest in object oriented programming which is very relevant to modern application development. Other relevant courses such as Software Engineering, Web Programming are included in the curriculum to help produce a more productive graduate. A number of foundation courses are included to make our graduates unique, all-rounded and critical thinkers.

There is a Research Project that is covered over three semesters. The Research Methods Course in second semester of second year provides the foundation on which students are expected to complete a successful research project. The Computer Science Project in third year, provides the basics and practice needed for students to specify their design, implementation and production of a complete project report which is assessed by a panel of Computer Science Department in the second semester of the final year.

Recess Terms or semesters, are not mandatory but they are available to provide students with practical skills in software engineering, systems engineering, software development, database programming and foreign languages (where it is possible).

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1.0 INTRODUCTION

1.1 Mbarara University of Science and Technology

Mbarara University of Science and Technology (MUST) was established in October 1989 by a Statute of the National Resistance Council, as the second public University in Uganda, after Makerere University. The Statute was repealed and replaced by the Universities and Other Tertiary Institutions Act 2001, as amended in 2003 and 2006.

The University was established in the former Nursing and Midwifery School and took over its facilities adjacent to the current Mbarara Regional Referral Hospital. It started with the pioneer Faculty of Medicine offering the degree of Bachelor of Medicine and Bachelor of Surgery (MBChB). The first cohort of students were 43 students, and 5 members of staff, four of whom were from the republic of Cuba. To date, the relationship with the Cuban Government has been maintained. Cuba provides at least 5 medical specialists to train students and also offer clinical services all the time.

The major aim of establishing the university was to bridge the gap of health professionals in the country. The University trains doctors and other health professionals such as nurses, pharmacists, medical laboratory scientists and recently, physiotherapist. These are trained to acquaint themselves with the rural conditions and health needs of the bulk of Uganda's population under the programme known Community Based Education Research and Services (COBERS).

The University continued to grow and new Faculties/Institutes were established. These include:

1. Faculty of Computing and Informatics
2. Faculty of Science
3. Faculty of Applied Science and Technology
4. Faculty of Interdisciplinary Studies
5. Faculty of Business and Management Sciences
6. Maternal New-born and Child Health Institute
7. Institute of tropical Forest Conservation

1.2 The University Motto

"Succeed We MUST"

1.3 Mission, Vision, Mandate and Objectives

Vision: To be a centre for academic and professional excellence in science and technology

Mission: To provide quality and relevant education at National and International level with particular emphasis on science and technology and its application to community development

Mandate: To Impart University education within Uganda with particular emphasis on scientific and technological education and their application to rural development

Objective

1. To advance, transmit and preserve knowledge from one generation to the next
2. To produce the necessary human resource in applied sciences, technology and management skills.
3. To generate and disseminate knowledge and innovation;
4. To provide services to the public in analysing and solving problems
5. To teach students to understand and appreciate local, national and international issues

1.4 Location

The University is located on two campuses;

1. Mbarara/Town Campus located on plot 8-18 Mbarara – Kabale Road equivalent to 21.45ha
2. Kahumuro Main Campus at 6 Kms, along Mbarara Fort Portal Road situated on 234.2ha of land at Kihumuro and Rukindo (which is across River Rwizi). The land at Rukindo is earmarked for agricultural and teaching activities.

1.5 The Faculty of Computing and Informatics

The Faculty of Computing and Informatics (FCI) established in 1997, under the Faculty of Science and shifted later to the Faculty of Development Studies as a department of Computer Science. In 2006, the department was elevated to the Institute of Computer Science. In 2016, the Institute was further transformed to the Faculty of Computing and Informatics which currently offers a dimension of programmes including – Bachelor of Computer Science, Information Technology, Computer Engineering, short courses and

postgraduate courses – MSc Information Systems and Health Informatics; Masters in Business Informatics among others. All these programs are aligned strategically with the University mission as providing quality and relevant education at national and international level with particular emphasis on Science and Technology and its application to community development. The FCI has fully established herself as a center of multidisciplinary research through diverse research areas in which all its staff are currently part of.

The Faculty has two academic departments;

1. Department of Computer Science
2. Department of Information Technology

1.6 The Department of Computer Science

The Department of Computer Science was established in 1997 under the Faculty of Science and later to the Faculty of Computing and Informatics, to run courses in Computing and technology in order to meet the increasing demand of computing and technology in the country.

The department aspires to be the university's great innovation centre recognized nationally and internationally in teaching and research. The Department of Computer Science now offers majorly undergraduate and postgraduate degree programs, and technical short courses to meet the demand for Computing and technology in the continent. The programs offered include;

1. Postgraduate Degree Program: Master of Science Information Systems
2. Undergraduate Degree Programs: Bachelor of Computer Science (BCS), and Bachelor of Science in Software Engineering

The Department also offers professional and technical short courses to meet the demand for computing and technology basic skills. These include: Certificate in Computer Applications (CCA), Cisco Certified Network, Associate (CCNA), Cisco Certified Network Professional (CCNP), Cisco Networking Academy Program (CNAP), Information Technology Essentials (ITE) I & II and International Computer Driving License (ICDL) Certification.

1.6.1 Infrastructure

The Department of Computer Science is currently located at the town campus in a space limited to the activities of the department. This space comprises of two-degree training computer laboratories, one short course technical laboratory and one practical training laboratory. The Department has three staff offices and one server room. The new structure housing over 15 office space has under construction at Kihumuro Campus – the main campus.

1.6.2 Human Resource

The Department of Computer Science has 7 PhD holders, and over 8 master's holders. Six of these staff are undertaking their PhD studies. There are also other members of staff undertaking their master's degrees. In particular, the department has over 10 members of staff who can be able to handle different courses in the department.

1.7 Rationale for Reviewing the Programme

The Department of Computer Science realized that there was a need for revising the curriculum for the Bachelor of Computer Science. This is because with technology changes and a dynamic Ugandan economy there is a need to develop curricula that meets these changing needs of the community. Therefore, a number of aspects fostered the review process:

- The former BCS was last peer reviewed in 2014. It is a requirement by National Council of Higher Education (NCHE) to review curricula after every five years.
- We held a workshop with industrial partners and alumni with the goal of obtaining feedback about the curriculum we were offering. The feedback obtained was important in the restructuring process of the curriculum.
- A number of current BCS students at the Department of Computer Science were also involved in obtaining feedback about the content that was being offered to them. The students helped us identify several areas of redundancy and duplication in the curriculum. Furthermore, they also helped in identifying particular courses that they would like to see included as part of the curriculum.
- We also considered the attractiveness and employability of CS graduates in Uganda. We considered the kind of jobs that were advertised for graduates in CS. This information was useful to help change the curriculum based on its strengths and weaknesses.

1.8 The Assessment Exercise

1.8.1 Description of the Assessment Process for Bachelor of Computer Science (BCS)

The Bachelor of Computer Science (BCS) curriculum was initially started in 1997 and accredited by National Council for Higher Education (NCHE). Mbarara University of Science and Technology having accepted and embraced quality assurance guidelines as prescribed by NCHE, the Faculty of Computing and Informatics (FCI) therefore undertook an assignment to review the programme, a course of action which is supposed to be done after every three years of programme execution. This process involved reviewing the opinions of key stakeholders in relation with the current market needs of the product. These including: lecturers, Heads of ICT Departments, current students, alumni, employers and government departments. The process further reviewed the programme rationale, objectives, learning

outcomes, structure, delivery modes and methods, assessment methods as well as the existing and recommended study materials and study support available.

1.8.2 Methods for Data Collection

The Dean together with the Head of Department Computer Science, and Quality Assurance (QA) coordinator were assigned the role of leading the process because of being representatives of the Faculty to the University Quality Assurance (QA) Committee. The questionnaires for the Self-Assessment Process (SAP) were sent out to key stakeholders. However, due to the busy working schedules of these two members, the entire department was involved. Meetings were organised chaired by the head of department, duties were assigned to both academic and administrative staff.

So far, all issues pertaining to this SAP exercise and to QA have been discussed and deliberated upon within the Faculty Board that acts as Faculty Quality Assurance Committee (FQAC). This is not to be considered necessarily as a permanent form of organisation and may be modified in future, by creating an ad hoc Committee, when the dimension of the Faculty and the increased number of staff members will make it advisable.

Quality Assurance in the Faculty has been following largely the resolutions of the University Senate as well as the guidelines from NCHE.

During several monthly meetings of the Faculty Board, the progress of the self-assessment exercise has been regularly analysed and discussed. The faculty intended to collect data from lecturers of the programme, the heads of ICT departments, current students, alumni, employers (labour market) who are key development partners and some government officials. We did not include the parents among the stakeholders; we thought the lecturers and the labour market would substitute them and give sufficient information to benchmark our reviewed curriculum. We also benchmarked from three International recognized institutions: 1). The University of Cape Town; 2). Hasso Plattner Institute (HPI) Germany; and 3). University of Nairobi. Table describes the respondents during the curriculum review exercise.

Table 1 – Distribution list of Questionnaires

Category of Respondent	No. questionnaires distributed (of potential respondents contacted)	No. of questionnaires returned filled in
Lecturers of the Programme	11	11
Head of ICT	3	2
Current Students	31	19
Alumni	29	17

Employers	5	2
Total	79	51

We experience major difficulties in the process of collecting data from our respondents most especially the continuing students whom we thought would respond and return all the questionnaires. Unfortunately, only 19 were returned.

Questionnaires for alumni and most employers were sent via e-mail and further contacts were made by telephone calls. Although many of the contacted persons expressed willingness and availability to answer the questionnaires, but the response was below average.

1.5 The Revised Bachelor of Science in Computer Science Program

The rapid growth of Information and Communication Technology (ICT) in Uganda has led to a need to keep updated CS professionals to be able to optimally move with the rapid emerging new technology.

The Bachelor of Science in Computer Science degree provides graduates with knowledge needed in computing related fields, and prepare each graduate for further advanced study in computing and related field. The programme exposes students to a solid foundation in AI, programming practices, cloud computing, Robotics, Systems and Networks Administration.

The course is a three-year programme with each year having two semesters. Each semester lasts fifteen (15) weeks, two (2) of which are for examinations. A total number of 40 course units totaling to 133 credit units are supposed to be realized by the time a student completes this programme

2.0 REQUIREMENTS, STAKEHOLDERS AND EXPECTED LEARNING OUTCOMES

2.1 Introduction

Today, ICT plays an increasingly important role in the dynamic growth of 21st Century organizations. Quick access to vast amounts of information determines the success of these organisations. Therefore, computer science has, as a result, become indispensable asset which one cannot do without.

In the year 1997, the institute of Computer Science initiated a Bachelor of Computer Science (BCS) under the Department of Computer Science, to fill this gap, which by then were in the field of computing.

In the year 2014, the Faculty reviewed the course to update it match with the new technologies at that time. Today (five years down the road), the Department is yet re-designing the same course to align it with the new requirements in the market and also as a requirement by the NCHE as mentioned earlier in the introduction.

The program emphasizes practical skills in the various aspects of computer science. The course also ensures that students get the necessary background knowledge on the subject matter covered. It further trains students to become competent in AI, programming practices, cloud computing, Robotics, Systems and Networks Administration among others.

Practical courses combined with solid theoretical knowledge and access to a large electronic library and various professionals from academia and industry ensure that graduates under this course are of world standard.

2.2 The Revised Curriculum

The Bachelor of Science in Computer Science is a three-year full time programme and one of the other programs offered in the department of computer science. The program is organized on a semester basis, with two semesters in any one academic year. At the end of the second year, during the long break, students are expected to take industrial/internship training exercise, a time when they are supposed to be exposed to real life experience and hand-on practice. The maximum period for the completion of the program is 4 years, except when the students has undergone cases of repeating having failed course(s), in which case the student is given up to 5 years; thereafter the student is requested to leave the University. The reviewed curriculum has put emphasis for content delivery on student centred learning methods. Despite the fact that they require limited number of students, these methods have been proved to promote student learning and academic achievement as well as enhancing students' satisfaction with their learning experience.

The body of knowledge for computer science is categorized in the frame of reference as follows:

- a) Preliminaries or Foundation courses;
- b) Pre-requisites course;
- c) Core courses;
- d) Electives; and
- e) Developmental or lifelong studies

2.2.1 The BSc CS Programme Rationale

The Department of Computer Science at Mbarara University of Science and Technology offers a unique computer science program designed to prepare students for a professional career in AI, programming practices, cloud computing, Robotics, Systems and Networks Administration, among others. The program was designed with the intention of making the program competitive while maintaining the high standard of the program. With the courses offered, the graduates will become highly competitive in the job market and at the same time prepare them in becoming independent in making choices of career and self-employment. Students are trained not only in the technical areas of specialization but also in personal development, communication skills and entrepreneurship.

2.3 The Programme Objectives

The programme aims at producing professionals who will be able to:

1. Establish successful careers in the field of computer science.
2. Explain strong foundation in computing and mathematical theoretical concepts, and be able to apply these concepts to problems requiring computer solutions.
3. Apply effective communication skills, and also team members and leaders that add value to employers and businesses regionally and globally
4. Stay current in emerging technologies through training, self-learning and/or graduate studies
5. Understand, and be proficient in the use of, system design notations and system design engineering process to develop software systems and solve problems in computer science
6. Explain ethical, legal and social issues faced in their workplace and contribute positively to the community

2.4 Requirements of stakeholders and expected learning outcomes

The requirements of stakeholders and expected learning outcomes are categorised into

1. Knowledge
2. Skills, and
3. Competencies

2.4.1 Knowledge

The learning outcomes for the BSc in Computer Science state that degree holders possess knowledge of:

1. Recurring themes, and a set of general principles that have broad application to the field of computer science
2. Social, legal, ethical, and cultural issues inherent in the discipline of computing
3. Software systems used in different domains. This requires both computing skills and domain knowledge
4. Software development fundamentals, including programming, data structures, algorithms and complexity
5. Systems fundamentals, including architectures and organization, operating systems, networking and communication, parallel and distributed computation, and security
6. Mathematics fundamentals, including discrete structures, statistics and calculus
7. Software engineering fundamentals, including software analysis and design, evaluation and testing, and software engineering processes
8. Application fundamentals, including information management and intelligent applications
9. Multiple programming languages, paradigms, and technologies

2.4.2 Skills

The learning outcomes for the BSc in Computer Science state that degree holders can apply the methods and procedures as follows:

1. Know how to apply the knowledge they have gained to solve real problems
2. Realize that there are multiple solutions to a given problem and these solutions will have a real impact on people's lives
3. Communicate their solution to others, including why and how a solution solves the problem and what assumptions were made
4. Successfully apply the knowledge they have gained through project experience
5. Encompass an appreciation for the structure of computer systems and the processes involved in their construction and analysis
6. Understand individual and collective responsibility and individual limitations as well as the limitations of technical tools
7. Understand the range of opportunities and limitations of computing

2.4.3 Competencies

The learning outcomes for the BSc in Computer Science state that degree holders can apply their knowledge and skills, as follows:

1. Understand the multiple levels of detail and abstraction

2. Recognize the context in which a computer system may function, including its interactions with people and the physical world.
3. Able to communicate with, and learn from, experts from different domains throughout their careers
4. Possess a solid foundation that allows and encourages them to maintain relevant skills as the field evolves
5. To be able to manage their own career development and advancement
6. Manage their own learning and development, including managing time, priorities, and progress
7. Have developed interpersonal communication skills as part of their project experience
8. Work effectively both individually and as members of teams
9. Make effective presentations to a wide range of audiences about technical problems and their solutions
10. Encompass an appreciation of the interplay between theory and practice

2.5 Admission Requirements

The general admission requirements shall be decided and revised periodically by the senate and made available by the office of the Deputy Vice Chancellor Academic Affairs (DVC AA); and/or academic registrar. The admission criteria of the BSc. CS shall consist of the following;

- i) Admission by Direct Entry Admission to the first year of the program will require a student to have sat the;
 - a) Uganda Advanced Level Certificate of Education (UACE) with at least a principal pass in mathematics plus a principal pass in any other subject(s).
- ii) Diploma Holder: A person who holds a diploma from STEM or its equivalent from a higher institution of learning recognized by NCHE and Mbarara University of Science and Technology (MUST).

2.6 Teaching Methods and Assessment

2.6.1 Mode of delivery

The BSc. CS programme will be conducted during the week from Monday to Friday starting from 8.00am to 5.00pm with a one-hour lunch break. The programme will employ a number of teaching and learning formats which include:

- a) Lectures, tutorial and practical sessions.
- b) In-class student presentations.
- c) Wide range of computer-based learning tools.
- d) The use of MUST LMS:

2.6.2 Course Assessments

The general assessment of courses in this programme will be based on hundred (100) total marks with proportions as:

- a) Continuous coursework consisting of 40 marks;
 - i. Class Assignments
 - ii. Mid Semester Tests
 - iii. Group course work and presentation
 - iv. Lab practical sessions
 - v. Seminars
 - vi. IT projects
- b) Examination consisting of 60 marks.

However, some courses have varying assessment distributions that are described in the detailed course descriptions. A minimum of two course assignments/tests shall be required per course.

2.6.3 Grading of Courses

The weighting unit is based on a Credit Unit (CU). A Credit Unit is one contact hour per week per semester or a series of seventeen (17) contact hours per semester. A contact hour is equal to (i) one lecture hour, or (ii) two practical hours. The grading of the program shall be as per the University policies and regulations appears in the hand book.

2.6.4 Calculation of Cumulative Grade Point Average (CGPA)

The Cumulative Grade Point Average (CGPA) shall be achieved by weighting each course, i.e., multiplying the grade point of the course by the number of its credit unit's value. The total of the grade points shall be divided by the total credit unit value as shown;

Equation 1 below shall be used for CGPA calculation:

$$CGPA = \frac{\sum_{i=1}^n (GP_i * CU_i)}{\sum_{i=1}^n CU_i} \quad \text{equation 1}$$

Where:

- *CGPA* – Cumulative Grade Point Average
- *GP_i* - Grade Point score of a course i (*GP_i* ranges from 0 to 5, see table 1);
- *CU_i* – Credit Units of course i (*CU_i* ranges from 3-5);
- *n* - The total number of courses so far accomplished.

2.6.5 Additional Grades

The following additional letters will be used, where appropriate:

1. W = Withdraw from Course;
2. I = Incomplete;
3. AUD = Audited Course Only;
4. ABS = Absent with Approval;
5. P = Pass;
6. F = Failure.

2.6.6 Pass Mark

A minimum pass grade for each course shall be 2.0 grade points.

2.6.7 Retaking a Course or Courses

1. A student shall retake a course(s) when next offered in order to obtain at least the pass mark (50%) if he/she had failed during the first attempt in the course(s). A student may take a substitute elective, where the Student does not wish to retake a failed elective.
2. A student who has failed to obtain at least the pass mark (50%) during the second assessment in the same course(s) he/she has retaken shall receive a warning.
3. Where students miss to sit examinations for justified reasons; they should not be recorded as those who retake when they sit the examinations when next offered.
4. A student shall not be allowed to accumulate more than five (5) retake courses at a time. Students are required to register for retake course(s) first before registering for new courses offered in that semester and the retake courses should fit into the approved normal load so as to avoid time table clashes.

2.6.8 Progression

Progression shall be regarded as normal, probationary or discontinuation as per the standard MUST Senate guidelines;

1. Normal Progress: This occurs when a student passes each course taken with a minimum Grade Point of 2.0.
2. Probationary: This is a warning stage and occurs if either the cumulative grade point average (CGPA) is less than 2.0 and/or the student has failed a core course. Probation is waved when these conditions cease to hold.
3. Discontinuation: When a student accumulates three consecutive probations based on the CGPA or the same core course(s), he/she shall be discontinued. A student who has failed to obtain at least the pass mark (50%) /grade point of 2.0 during the third assessment in the same course(s) he/she had retaken shall be discontinued from his/her studies at the University.

2.6.9 The Computing Research Project

Students are required to demonstrate their ability to independently formulate a detailed project proposal, as well as develop and demonstrate their project thoroughly.

1. A candidate shall submit a project proposal to the Department of Computer Science research Committee during the second semester of year two.
2. A candidate shall be allowed to formally start on the project after the second semester.
3. A candidate shall be assigned a supervisor who is a specialist in the candidate's field of study to undertake supervision of the project.
4. The candidate shall execute the project during the third year.
5. The candidate shall submit the project report by the end of the third year.

To pass the project, the candidate shall satisfy the examiners in a written report and viva-voce independently.

2.7 The Mbarara University of Science and Technology Service Courses

Few universities in the region emphasize a holistic education. MUST believe in an education that integrates physical, social, emotional, and economic growth with traditional academics. For example, all students will take "Development Studies" as service course in Political Economy, and Citizenry, Professionalism, Globalization and Entrepreneurship. The University's goal in teaching is to facilitate learning, not simply to help students pass exams. To accomplish this, MUST education philosophy limits the number of courses a student can take in a semester to promote depth of study. Practical experience is required in many programmes.

2.8 Knowledge Areas

The skills and competencies of computer science graduates can be encompassed in eighteen knowledge areas which are;

1. Algorithms and Complexity (AL)
2. Architecture and Organization (AR)
3. Computational Science (CN)
4. Discrete Structures (DS)
5. Graphics and Visualization (GV)
6. Human-Computer Interaction (HCI)
7. Information Assurance and Security (IAS)
8. Information Management (IM)
9. Intelligent Systems (IS)
10. Networking and Communications (NC)
11. Operating Systems (OS)
12. Platform-based Development (PBD)
13. Parallel and Distributed Computing (PD)
14. Programming Languages (PL)
15. Software Development Fundamentals (SDF)
16. Software Engineering (SE)
17. Systems Fundamentals (SF)
18. Social Issues and Professional Practice (SP)

2.9 Target Jobs

From games developer to manager of IT and communications services, the computer scientist will have a range of opportunities open to them as computer science graduate. These include among others;

1. Application analyst
2. Applications developer
3. CAD technician
4. Cyber security analyst
5. Data analyst
6. Database administrator
7. Forensic computer analyst
8. Game designer
9. Games developer
10. Information systems manager
11. IT consultant
12. Machine learning engineer
13. Multimedia programmer

14. Penetration tester
15. SEO specialist
16. Software engineer
17. Systems analyst
18. UX designer
19. VFX artist
20. Web designer
21. Web developer

2.10 Opinion

The BSc CS strives to fill in the constantly evolving need for IT professionals in Uganda by producing graduates that are able to identify and realize current emerging needs of the stakeholders (industry, alumni, students, and organizations) as reflected the learning outcomes summarised in section 2.4.

3.0 THE PROCESS

3.1 Program specification

The courses specification consists of the following:

- i. **Course description** – What the course is all about. It may contain a general statement for the program outcome, superficially mentioning the program contents.
- ii. **Course objectives** – Why is the course important, why is the University having this course in the program; and many of the facets of the course that render it useful to the society and the world of work?
- iii. **Learning outcome** – Identify what the learner will know and be able to do by the end of a course or program.
- iv. **Methods of Delivery** - Define how course offerings are presented to learners
- v. **Course content** – This details the materials to be studied for each course in the program.

3.2 Program Content

The detailed programme content, spelt out course units offered in every semester for six semesters that we have in three years of study. The timetable is prepared for every semester and mentioned early, that guides the delivery of the programme. The Table below shows the program content.

Curriculum Structure

CURRICULUM FOR BACHELOR OF SCIENCE IN COMPUTER SCIENCE									
Code	Course Name	Type	LH	TH	PH	CH	CU		
Year 1 Semester 1									
CSC1101	ICT Principles	C	45		30	60	4		
CSC1102	Principles of Programming	C	45		30	60	4		
CSC1103	Computer Organization and Architecture	C	45		30	60	4		
MTH1101	Discrete Mathematics	C	30	30		45	3		
MTH1102	Differential Calculus	C	30	30		45	3		
DVS1106	Introduction To Political Economy	C	30	30		45	3		
CSC1104	Communication and Writing Skills for Computer Scientists	C	30	30		45	3		
Required Total Credits							24		
Year 1 Semester 2									
CSC1201	Computer Networks	C	45		30	60	4		
CSC1202	Object Oriented Programming I	C	45		30	60	4		
CSC1203	Data Structures and algorithms	C	45		30	60	4		
CSC1204	Database Systems	C	45		30	60	4		
MTH1201	Applied Linear Algebra	C	30	30		45	3		
MTH1202	Integral Calculus	C	30	30		45	3		
DVS1203	Political Economy of Africa And Development	C	30	30		45	3		
Recess Term I Electives									
CS1206	Networks Certification								
Required Total Credits							25		
Year 2 Semester 1									
CSC2101	Operating Systems	C	45	30		60	4		
CSC2102	Object Oriented Programming II	C	45		30	60	4		
CSC2103	Web Programming	C	30		30	45	3		
CSC2104	Database Programming	C	45		30	60	4		
CSC2105	Systems Analysis and Design	C	30		30	45	3		
CSC2106	Scientific Thinking	C	30	15	15	45	3		
MTH2101	Differential Equations	C	30	30		45	3		
DVS2105	Political Economy of Uganda	C	30	30		45	3		
Required Total Credits							27		
Year 2 Semester 2									
CSC2201	Software Engineering	C	30	30		45	3		
CSC2202	Networks and Systems Administration	C	45	0	30	60	4		
CSC2203	Information and Cyber Security	C	30		30	45	3		
CSC2204	Research Methods in Computing	C	30		30	45	3		

CSC2205	Computer Graphics	C	30		30	45	3	
DVS2201	Citizenry, Professionalism, Globalization And Entrepreneurship	C	30	30		45	3	
	Elective (Choose one)							
CSC2301	Recess Semester: Internship	C	0	0	120	60	4	
Electives (Choose 1)								
CSC2207	Web Application Development	E	30		30	45	3	
CSC2208	Distributed and Cloud Computing	E	30		30	45	3	
Required Total Credits							26	
Year 3 Semester 1								
CSC3101	Software Project Management	C	30		30	45	3	
CSC3102	Artificial Intelligence	C	45		30	60	4	
CSC3103	Embedded Systems	C	45		30	60	4	
CSC 3104	Computational Complexity	C	30		30	45	3	
Electives (Choose 1 ratio 4:1)								
<i>Systems & Networks Electives</i>								
CSC3104	Multimedia Technology	E	45		30	60	4	
CSC3105	Wireless and Mobile Communications	E	45		30	60	4	
CSC3109	Mobile Applications Development	E	45		30	60	4	
CSC3111	Computer Vision & Image Processing	E	45		30	60	4	
Required Total Credits							18	
Year 3 Semester 2								
CSC3201	Business and Entrepreneurship	C	30		30	45	3	
CSC3202	Computer Science Project	C			120	60	4	
CSC3203	Compiler Construction	C	30		430	45	3	
Electives (Choose 1, ratio 3:1, Major: Minor)								
<i>Systems & Networks Electives</i>								
CSC3205	Parallel Processing	E	30		30	45	3	
CSC3206	UNIX Shell programming	E	30		30	45	3	
<i>Software Engineering Electives</i>								
CSC3209	Design Principles of HCI	E	30		30	45	3	
CSC3210	Data Science	E	30		30	45	3	
Required Total Credits							13	

Total Credits: 133

KEY

C: Core Course Unit

E: Elective Course Unit

CS: Computer Science

4.0 DETAILED COURSE DESCRIPTIONS

4.1 Year 1 Semester 1

Course Name	ICT Principles
Code	CSC1101
Type	Core

Level	Year 1 Semester 1
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Course Description

This course introduces the field of computer science and the fundamentals of computing including introductory topics, internet and networks, graphics and web programming (HTML 5, CSS and JavaScript). Introduction to computing is specifically designed for students with no prior programming experience, and taking this course does not require a background in Computer Science. The course enlightens students on the nature and uses of computers, providing them with the necessary Knowledge & Skills set to work effectively in computing environment.

Course Objectives

- i. Students will learn what computer science is about, especially data representations, algorithms, encodings, forms of programming.
- ii. To provide a sense of background, history, and origins of internet and communication technologies and justify the evolution of the new communication technologies.
- iii. To have students engage with technology in a series of discussion and hands-on exercises (where possible) with reasonable awareness of the ways in which communication technology is practically used in work related situations.
- iv. Students will be able to do Image manipulation using Adobe Photoshop, sound synthesis and editing, text (e.g. HTML) creation and manipulation.

Course Learning Outcomes

At the end of the course, students are able to;

- i. Explain basic ICT concepts.
- ii. Explain general computer organisation.
- iii. Identify and Explain the Computer Generation and their major technologies
- iv. Explain the different operating system concepts
- v. Identify types of Networks, Network Topologies and Networking Devices.

Course Outline

Topic	LH	TH	PH
Basic concepts of computer science	2	1	2
Internet and communication technology	2	1	2
Word processors, spreadsheets & Presentation Skills	2	1	2
Database Basics	2	1	2
Computer Graphics (Adobe Photoshop)	2	1	2
Audio and video editing	2	1	2
HTML and HTML 5	4	2	4
CSS	4	2	4
Introduction to JavaScript – Dynamic Web Pages, HTML Event Handlers, the on click Event Handler, Interaction via Buttons, Input Buttons, Dynamic Text within a Page, Alert Windows, The inner HTML Attribute – User Input via Text Boxes, The Text Box Element, JavaScript Variables, JavaScript Numbers and Expressions, Text Boxes and parseFloat, Programming Errors and Debugging	4	2	4
Total CU	24	12	24

Mode of Delivery

- Lectures: 24 hours
- Tutorials: 12 hours
- Practical: 24 hours

Mode of Assessment

- Coursework 40%
 - Programming project - 15%
 - Problem sets - 15%
 - Tests - 10 %
- Exam - 60%

Basic Reading List

- David Reed, *Balanced Introduction to Computer Science*, A, 3rd Edition, Pearson, 2011

- Tannenbaum A. S., Computer Networks, ISBN: 0133942481, Prentice Hall. Pearson, Information Technology, 3rd Edition Prentice Hall 2004 Kurose, J.F and Ross, K.W., Computer Networking: A Top-down Approach Featuring the Internet; ISBN: 0-201-47711-4; Addison-Wesley Longman.
- Van Steen, M., Sips, H., Computer & Network Organization, Prentice-Hall.
- Stallings William, Network & Internet work Security Principle & Practice, Prentice Hall. Hodson Peter Local Area Networks, Continuum Publishers
- David Flanagan, *JavaScript: The Definitive Guide*, 6th Edition, O'Reilly, 2011

Course Name **Principles of Programming**
Code CS1102

Prerequisite None
Level Year 1 Semester 1

Course Description

This course is an introductory programming course in which students learn basic and essential concepts of programming using C/C++. Topics to be covered in this undergraduate level course include language syntax, data types, the concept of variable scope, arrays, pointers, structures, objects, functions and function call structure, parameter passing, and the sequence, selection (if-then-else), and repetition control structures (for, while, repeat-until loops). We will also discuss introduction to Object Oriented design strategies and features of C++ that support OOP. No prior knowledge of C/C++ is required but students are expected to have a good understanding of Computing environment or concepts of Computer Literacy.

This course will in turn prepare students for programming in Visual C++, Visual C#, Java, Perl, Ruby on Rails, Python or any other language such as JavaScript or PHP.

Course Objectives

At the end of this course, students will:

1. Understand basic skills needed in programming
2. Write, compile, debug and run a program in C/C++
3. Understand the uses of all data types in C and will be able to declare data variables of all types and constants in a program.
4. Understand the use of functions and write functions in C/C++
5. Explain the use of Arrays and pointers and can use them in a program.

Course Learning Outcomes

At the end of the course, students should be to;

1. Discuss basic concepts of computer programming and developer tools.
2. Discuss the syntax and semantics of the “C/C++” language as well as data types offered by the languages
3. Demonstrate skills of how to write programs using standard language infrastructure regardless of the hardware or software platform
4. Demonstrate skills of code abstraction and composition in building large “C/C++” programs

Course Outline

TOPIC	LH	TH	PH
Introduction to Programming, Getting Started with C/C++, Exploring the IDE	2	2	2
Variables and Arithmetic Expressions, Symbolic Constants, Character Input and Output, File Copying, Character Counting, Line Counting, Word Counting	2	2	2
Types, Operators and Expressions Variable Names, Data Types and Sizes, Constants, Declarations, Arithmetic Operators, Relational and Logical Operators, Type Conversions	2	2	2
Operators and Expressions, Conditional Expressions, Precedence and Order of Evaluation, Input and Output, Formatted input and output	2	2	2
Control Flow, Statements and Blocks, If-Else, Else-If, Switch	2	2	2
Loops - While and For, Loops - Do-While, Break and Continue, Goto and labels	2	2	2
Functions and Structures	2	2	2
Pointers and Arrays, Character Pointers and Functions, Pointer Arrays; Pointers to Pointers	2	2	2
Multi-dimensional Arrays, Pointers vs. Multi-dimensional Arrays, Command-line Arguments, Pointers to Functions	2	2	2
Introduction to OOP using C++	2	2	2
Total	20	20	20
Total CH	60		

Mode of Delivery

- Lectures
- Tutorials
- Practical

Mode of Assessment

- 3 written tests will be administered to students.
- 1 Semester-Long- Challenge Project will be administered and this will capture all concepts learned in class
- Practical exercises will be done weekly

Assessment Tool	Grade Breakdown
Tests	10%
Practical Exercises	15%
Semester-Long- Challenge Project	15%
Final Exam	60%

Basic Reading List

1. Walter Savitch, *Problem Solving with C++*, 10th Edition, Pearson, 2018
2. Brian W. Kernighan, Dennis M. Ritchie, *The C Programming Language*, Second Edition.
3. Deitel H. M and P. J Deitel (2004). *C How to Program*, Fourth Edition, New Jersey: Prentice-Hall.
4. Gottfried, Byron S. (1996). *Schaum's outline of programming with C*, Second Edition, New Delhi: McGraw Hill.
5. Tony Gaddis, *Starting Out with C++*, Fifth Edition
6. Timothy B. D'Orazio, *PROGRAMMING IN C++*, ISBN 0-07-242412-5

Course Name **Computer Organisation and Architecture**
Code **CSC1103**

Level **Year 1 Semester 1**

Course Description

This course is concerned with developing an understanding of the different components of a digital computer, their individual operation/function and their organization as a complete unit. Emphasis will be placed on the major component subsystems of high performance computers: memory hierarchies' CPU structure and organization, Data Representation, input/output. An abstract view of this organization with its capabilities and programming model usually referred to as its architecture will also play a big part in helping students understand the computer

Course objectives

1. Explain the overview of computer organization and architecture through knowing the structure and function of computers
2. Have detailed knowledge about the major computer components, their functionality and the interconnection mechanism that enables them to communicate.
3. Understand the central processing unit (structure and function) and all its major components and how they interact to execute instructions, hence should be well versed with computer arithmetic, instruction sets, and control unit operation.
4. Understand digital logic

Learning Outcomes

At the end of the course, students should be to;

1. Have detailed knowledge about the major computer components, their functionality and the interconnection mechanism that enables them communicate
2. Know the input-output module of a computer enabling input of data and output of information.
3. Explain how the peripheral devices get connected to computer.
4. Understand the Central Processing Unit (structure and function) and all its major components and how they interact to execute instructions, hence should be well versed with computer arithmetic, instruction sets, and control unit operation.
5. Understand digital logic that make the computer a digital machine.

Course outline

Topic	LH	TH	PH
Overview of computer organisation and architecture	4	2	4
The Computer System	4	2	4
Arithmetic and logic	4	2	4
The central processing unit	4	2	4
Parallel organization	4	2	4
The control unit	4	2	4
Total	24	12	24

Total CH		60
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Course Content

Overview of computer organisation and architecture

- Basic Concepts and Computer Evolution
- Performance Issues

The computer system

- A Top-Level View of Computer Function and Interconnection
- Cache Memory
- Internal Memory Technology
- External Memory
- Input/Output
- Operating System Support

Arithmetic and logic

- Number Systems
- Computer Arithmetic
- Digital Logic

The central processing unit

- Instruction Sets: Characteristics and Functions
- Instruction Sets: Addressing Modes and Formats
- Processor Structure and Function
- Reduced Instruction Set Computers (RISCs)
- Instruction-Level Parallelism and Superscalar Processors

Parallel organization

- Parallel Processing
- Multicore Computers
- General-Purpose Graphic Processing Units

The control unit

- Control Unit Operation
- Microprogrammed Control

Mode of delivery

- Lectures Hours
- Tutorial Hours
- Practical Hours

Mode of assessment

- Continuous assessment (Take Home Assignments, practical and written tests) = 40%
- Final Examination = 60%

Basic Reading List

1. D. Patterson and J. Hennessy, *Computer Organization and Design: The Hardware/Software Interface*. Morgan Kaufmann Publishers, Inc. ISBN 1558606041, (2004-08-02).
2. William Stallings, *Computer Organization and Architecture*, 10th Edition, Pearson, 2016
3. Gary Michael Schneider (1985). *The Principles of Computer Organization*.
4. David Tarnoff: *Computer Organization and Design Fundamentals* E-Book (Pub.www.lulu.com 2007).

Course Name	Discrete Mathematics
Code	MTH1101

Level	Year 1 Semester 1
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Course Description

The goal of this course is to introduce students to ideas and techniques from discrete mathematics that are widely used in science and engineering. The primary themes of this course are the notions of proof, recursion, induction, modelling and algorithmic thinking. Logic and set theory are developed in this course and are then used throughout, to provide the preciseness of thinking that is emphasized. Key topics involving relations, graphs, trees, and formal languages and computability are covered in this course. In this course, the student learns to see discrete mathematics as a mathematician or theoretical computer scientist, to understand what is important and why. This course teaches the students techniques in how to think logically and mathematically and apply these techniques in solving problems. To achieve this goal, students will learn logic and proof, sets, functions, as well as algorithms and mathematical reasoning.

Course Objectives

The objectives of the course are:

- To present the basic concepts of discrete mathematics and ways of thinking that are especially important in information technology, computing and other disciplines.
- To raise awareness that a relatively small number of basic Mathematical concepts are applied to a wide variety of situations
- Understand and construct mathematical arguments.

Learning Outcomes

Upon successful completion of this course students should be able to: -

- Determine the size of a set without explicitly counting its elements
- Explore combinatorial identities
- Use graph theory to model real life situations

Course Outline

Topic	LH	TH
Logic and Proofs	4	1
Basic Structures	4	1
Algorithms, the Integers and Matrices	4	1
Induction and Recursion	4	2
Counting	2	1
Relations	2	1
Graphs	4	2
Trees	4	2
Modelling Computation	4	2
Total	32	13
Total CH	45	

Mode of delivery

- Lectures
- Tutorials

Mode of Assessment

There are three one-hour tests, each based on the topics for three assignments. Success in the tests is therefore highly dependent on how seriously you work at the weekly problems.

- 40% - Coursework mark
- 60% - Examination mark

Basic Reading List

1. Kenneth Rosen and Jerrold Grossman, *Discrete Mathematics and Its Applications*, 7th Edition, 2011
2. Jerrold W. Grossman, *Discrete Mathematics: An Introduction to Concepts, Methods, and Applications*, Macmillan 1989

Course Name	Differential Calculus
Code	MTH 1102

Level	Year 1 Semester 1
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Course Description

This course is designed to provide students with the foundational mathematical skills involving differential calculus that they will need in their major programs. Topics include: an in-depth study of limits of a variety of functions; the continuity of functions; derivatives, and techniques to compute them; and applications of derivatives including optimization problems. Emphasis will be placed on both acquiring skills and on gaining a deeper understanding of the concepts behind the techniques and algorithms used. Calculus is an immensely rewarding subject to study: emanating from centuries of human thought, it enlightens a broad range of modern human activity. The course will be demanding and I expect you to put many hours of effort to learn the material. I will do my best to guide you through the fundamentals of the subject, but, ultimately, the responsibility to learn the material, ask for help when you need it, and assess your progress rests with you.

Course Objectives

The objectives of the course are:

1. Explore fundamental concepts of single variable calculus
2. Explore solutions of problems from a mathematical perspective,
3. Prepare students to succeed in upper level math, science, engineering and other courses that require calculus.

Learning Objectives

By the end of this course, students will be able to;

1. Evaluate the limit and continuity behaviours of a given real-valued function
2. Apply this evaluation to describe and graph these functions
3. Appreciate the analytic and geometric ideas of differentiation
4. Compute the derivatives of a broad range of functions
5. Model and solve basic problems including those arising in biology (growth and decay), chemistry (mixing problems), and physics (related rates, electric circuits, work)

Course Outline

Topic	LH		TH
Angles; trigonometric functions and identities	2		1
trig. (in)equalities; one-to-one functions; inverse trig functions; graphs of second degree functions; functions review	2		1
exponential, inverse, and logarithmic functions; introduction to limits	2		1
using limit laws; limits at infinity; asymptotes; introduction to continuity	2		1
continuity; tangent lines; introduction to derivatives; derivatives of polynomial and exponential functions	2		2
product and quotient rules; derivatives of trigonometric; the chain rule	4		2
implicit differentiation; derivatives of logarithmic functions; exponential growth and decay	4		2
- related rates; linear approximations and differentials; hyperbolic functions - Newton's method; antiderivatives	4		2
- maximum and minimum values; the mean value theorem; how derivatives affect the shape of a graph - L'Hospital's rule; curve sketching; optimization	4		2
catch-up and review	4		
Total CU	45		

Mode of Delivery

- Lectures
- Tutorials

Mode of Assessment

- Coursework 40%
- Exam - 60%

Basic Reading List

1. Thomas, Calculus: Early Transcendentals, 13th ed. MyMathLab is required and contains an electronic version of the textbook. Access codes and registration information are available on our course t-square page. You can access our MyMathLab course at mymathlab.com. We will cover many of the topics in chapters 1-5.

COURSE NAME:	Introduction to Political Economy
CODE:	DVS1106
COURSE LEVEL:	Degree, Year 1 Semester 1
CREDIT UNIT:	3

Prerequisite: None

Purpose/Aim: The course aims at helping students understand the nature and subject matter of Political Economy and make them appreciate the laws that govern the development of societies.

Description: Political economy is concerned with the interaction of political and economic processes in a society i.e. the distribution of power and wealth between different groups and individuals, and the processes that create, sustain and transform these relationships over time. The ways in which allocation and distribution of resources is done normally change over time, and therefore the course focuses on the laws that govern the development of the society, economy and power relations and various institutions that support them at each stage of human development and the subsequent social conflicts and institutions that are created to contain them. Political economy makes extensive and intensive use of class analysis in making sense of society and history, but does so in the context of political, cultural, and environmental processes, as well as other economic processes. It is concerned with understanding the role of various social economic systems in shaping and transforming society and history. The interdisciplinary nature of Political Economy is designed to offer students an exposure to the core concepts of political economy. The course focuses on

investigating the relationship between the economy and politics as they affect the social and cultural life of societies, and secondly, focuses on the historical processes whereby social change is located in the interaction of the economic, political, cultural, and ideological moments of social life.

Objectives:

1. Understand the meaning and concerns of Foundations of Political Economy
2. Understand the various concepts associated with political economy
3. Understand the relations between economy and political power and the ways in which economic policy are shaped by political institutions
4. Understand the laws governing the development of societies

Course content:

Introduction

Hours

- Key terms in Political Economy, 3
- Historical development of political economy 3
- Why study Political Economy? 3
- Methodological approach to the study of society 3
- Hegel Philosophy & Karl Marx's Historical and Dialectical Materialism 3

Modes of Production

- Primitive Communalism 1
- Slavery 2
- Feudalism 2
- Capitalism- Mercantilism, Industrial Capitalism, Imperialism 3
- Socialism-Socialism, Social Democracy and Democratic Socialism 3
- Communism 1

Socio-economic systems

- Planned/Command Economies 3

Free Market/Capitalist Economies 3

Mixed Economies 3

Introduction to Theories of International Political Economy-

- Realism 3
- Liberalism 3
- Marxism 3

No.	COURSE OUTLINE	LH	PH	TH	CH	CU
						3
1.	Introduction to Political Economy	15	0	0	15	
2.	Modes of Production	12	0	0	12	
3.	Socio-Economic Systems	9	0	0	9	
4.	Introduction to Theories of International Political Economy	9	0		9	
5.				0		
6.				0		
7.				0		
	Total CU	45			45	3

Mode of delivery:

- Lectures
- Case studies and class exercises.
- Group assignments and presentations.
- Group reading.

Mode of Assessment:

No	Mode of Assessment	Mark (%)
1	Mid-semester exam or take home assignment	30
2	Class attendance and Students presentations	10

3	Final Examinations	60
	TOTAL	100

Instructional Materials

1. LED Projectors
2. Laptops
3. Teaching boards and writers
4. Rooms with enough chairs and tables.

Learning outcomes

At the end of the course, students will be expected to:

1. know relations between economy and political power and the ways in which economic policy are shaped by political institutions
2. know that there are laws governing the development of societies
3. Comprehend the transformation processes that are embedded in the society transformation from socio-economic system to another and understand the basics of conducting class analysis
4. Appreciate the dynamics of development issues today and as future policy makers, start to formulate their own views

Reading materials:

1. Heywood Andrew. (2000). *Key Concepts in Politics*. New York: Palgrave
2. Heywood Andrew. (2002). *Politics*. New York: Palgrave
3. Sen Gautam. (2005). "The political economy of Development" in Haynes, J.,ed (2005) *Development Studies*. England: Palgrave Macmillan pp 89 – 115
4. Wadada Nabudere. (1978). *The Political Economy of Imperialism*. Dar es Salaam: Tanzania Publishing House
5. Bowles Smauel, Richard Edwards, Frank Roosevelt. (2005). *Understanding Capitalism, Competition, Command and Change*. New York and Oxford: Oxford University press.
6. Cutler Anthony, Hindess Barry, Hirst Paul & Hussain Athar. (1977). *Marx's Capital and Capitalism Today*, Vol I. London: Routledge & Kegan Paul

7. Clilote R. (1990). *The Political Economy of Imperialism: Critical Appraisals*. Kuwer: Academic Publishers.
8. Nabudere, D. W. (1980). *Imperialism and Revolution in Uganda*. Dar es Salaam: Tanzania Publishing House
9. Mandel Ernest, 4th Edition. (1977). *Marxist Economic Theory*. London: The Merlin Press

Course Name	Communication and
Code	Writing Skills for
	Computer Scientists
	CSC 1104

Level Year 1 Semester 1

Course Description

The course covers material that will develop the students' understanding of communication both conceptually and practically. The course aims to provide students with a strong framework for the development of practical communication skills in organization settings. One major aim of this course is to stimulate the student to think about what, why and how to communicate. Another is to develop a critical understanding of communication theory, process and techniques at organizational, managerial and personal levels. This is achieved by evaluating theories and experiential activities. Through the process the student should become a more effective communicator in both formal and informal communication settings.

Course Objectives

1. Analyze and assess effective human communication
2. Develop and practice human communication skills, including skills in verbal, non-verbal, listening, critical thinking, small group and public speaking
3. Understand effective human communication techniques and processes

Learning Outcomes

Upon successful completion of this course, students should be able to:

1. Describe the communication process.
2. Use listening skills and questioning techniques and identify and minimize influencing factors.
3. Use communication tools and communicate in the workplace.

4. Apply scientific leadership skills to work activities.
5. Communicate effectively both verbally and non-verbally.
6. Apply the requisite academic communication skills in their essay writing and other forms of academic writing.
7. Demonstrate understanding of the generic fundamentals of communication.
8. Use computer-mediated communication platforms in academic and professional work.
9. Relate to the interpersonal and organizational dynamics that affect effective communication in organizations.

Course Content

NO	TOPICS	HRS
1	Fundamental Skills including reading, listening, note-taking, and note making, speaking and interacting skills.	10
2	Interpersonal Skills will cover conduct and interaction at the work place, conduction of meetings, writing an agenda and a notice of a meeting, deciding on items for discussion, organizing the agenda, inviting people for a meeting and writing of minutes.	15
3	Technical writing will include: labeling and specification formats and other technical documentations. Academic writing will address report writing, seminars and workshop paper presentation; inter-office and intra-office communication with particular emphasis on business correspondence and memo writing.	10
4	Public Speaking and Presentation skills • Effective public presentation skills • Audience analysis • Effective argumentation skills • Interview skills	10
Total CU		45

a) Mode of Delivery

Lectures, Group Discussions, assignments and online discussions

b) Teaching and Learning Resources

Text books, Journals and publications, computers and power point projectors.

c) Mode of Assessment

Unit	Score (%)
Tests	20
Assignments	10
Case studies	10
Final examination	60
Total	100

Recommended Reading Books

1. M. McKay PhD and M. Davis PhD, *Messages: The Communication Skills Book*
2. Mar 3, 2009
3. N. A. Saleemi, *Business Communication and Report Writing Simplified*, 1st ed., N. A. Saleemi Publishers.
4. S. S. Nezamdoost, I Want to Talk!: *Social and Behavioural Sciences* 2014, Vol.98, pp.1346-1355
5. Martha Davis, Ph.D. Paleg. Kim, and Patrick Fanning, *The Messages Workbook: Powerful Strategies for Effective Communication at Work and Home*, Nov 2004
6. J. T. Wood, *Communication Mosaics: An Introduction to the Field of Communication*, 2013
7. Fielding, M. (1997) *Effective Communication in Organisations* 2nd ed. Kenywyn: Juta +Co
8. Jandt, F.E. (2004) *An Introduction to Intercultural Communication: Identities in a Global Community* 4th ed. London: Sage Publications.
9. Lucas, S.E. (2004) *The Art of Public Speaking* 8th ed. McGraw Hill
10. DiSouza, J.R and Leggie N.J. (2000) *Business and Professional Communication* 2nd ed. Kenywyn: Juta +Co.

Year 1 Semester 2

Course Name	Computer Networks
Code	CSC1201
Type	Core

Prerequisite	None
Level	Year 1 Semester 2

Course Description

Computer Networks equips students with techniques for building and managing data and computer networks. It gives the opportunity for undergraduate students to have a basic understanding of computer networks which seem to be the fastest growing technologies in our society today. Computer networks has led to detailed research on the underlying structures, interfaces and data transmission circuits that enable information exchange. Today we see networks as the building blocks upon which we can base today and tomorrows ICT industry.

This course will therefore introduce the concepts, theory, practice, design and organisation associated with computer networks; it will also provide a foundation for further studies in this study area.

Course objective

This course aims at using lectures, exploratory reading and labs, to enlighten bachelors' students of IT on aspects concerning computer networks; providing them with the necessary Knowledge & Skills set to work effectively in a network oriented environment.

Learning Outcomes

Upon completion of this course, students should be able to:

- a) Appreciate and demonstrate knowledge on contemporary issues in computer networking
- b) Understand, develop, specify, assemble, assess and manage data and telecommunication systems, networks and interfaces for institutions/companies that employ telecommunication.
- c) Appreciate and apply hardware and software system design principles and diagnostic methodologies in building computer networks.
- d) Install systems required for their employer's business, while adapting to rapid changes both in the technology used to implement advanced computer networks and their services.

Course Outline

Topic	LH	TH	PH
Introduction to computer Networks; Types and uses of computer networks and communication Models	2		
Network Software; Protocol hierarchies, design issues for layers, connectionless oriented services, service primitives, the relationship of services to protocols.	4		4
Data Transmissions; Concepts and terminology, Analog and digital transmission, Transmission impairments and channel capacity.	4	2	4
Transmissions media; Guided transmission media, wireless transmission and propagation and line of sight transmission.	4	2	2
Digital data Communication Techniques; Asynchronous and synchronous transmission, Errors-types, Detection and correction, Line configuration, Interfacing, Multiplexing and spread spectrum	4	2	2
LANs; Topologies, Choice of topology and transmission media, LAN Devices; Bridges, Hubs, Switches, Repeaters.	4	2	6
WANS; Switching-circuit and packet switching, X.25, Frame relay, ATM and Introduction to Routing	4	2	6
Total	26	10	24
Total	60		

Mode of delivery

Lectures
Tutorials
Practicals

Mode of assessment

Coursework (tests and take home assignment) = 40%
Final examination=60%

Basic Reading List

1. Andrew S Tanenbaum, Computer Networks, 4th Edition, NJ: Pearson Education International, ISBN: 0-13-038488-7
2. **D. Russell**, The principles of computer networking, UK-NY: The press syndicate of the University of Cambridge, 1997.
3. **Wendell Odom**, Computer Networking First-Step, Cisco Press, United states of America, 2005, ISBN: 1-58720-101-1.
4. **Kumar Ajit Singh**, Computer Networks, Firewall media, New Delhi

5. **William Stallings**, Data and Computer Communications, 7/8 Edition, New Delhi: Prentice-Hall of India.
6. **Behrouz A. Forouzan**, Data Communications and Networking, 4th Edition, NY: Mc Graw Hill, 2007.
7. **Fred Halsall**, Data Communications, Computer Networks and Open Systems, 4th Edition.
8. **Douglas E. Comer**, Internetworking with TCP/IP, Principles, Protocols, and Architecture, 5th, Edition.
9. **James F. Kurose and Keith W. Ross**. Computer Networking - A Top Down Approach Featuring the Internet, 3rd edition, Addison-Wesley, ISBN 0-321-22735-2.
10. **Christopher E. Strangio**, Data Communications Basics, A brief Introduction to Digital Data Transfer, CAMI Research Inc, Acton, Massachusetts, 2006.

Course Name	Object Oriented Programming 1
Code	CS1202
Type	Core
Contact Hours	45
Credit Units	3
Prerequisite	None
Level	Year 1 Semester 2

Course Description

This course is designed for undergraduate students in computer science and computational science who need to learn fundamental programming skills. The purpose of this course is for students with little or no background in programming methods to be introduced to the fundamental computer programming. Due to the nature and complexity of this field of study, the course will be time consuming and requires serious dedication on the part of each student. Among other topics, students learn how to solve problems, transform the solution into a program written in java programming language, and vice versa. Topics include control structures, arrays, functions, classes, objects, file handling, and simple algorithms for common tasks.

Course Objectives

At the end of this course, students will:

1. learn how to solve problems, transform the solution into a program written in the Java programming language
2. Have basic knowledge to apply with other technologies. Existing and future technologies
3. Know how to manage a project
4. Be innovative thinkers & be able to apply the technologies with real business

Learning Outcomes

On completion of the course students should be able to:

1. Understand the features of C++ supporting object oriented programming
2. Understand the relative merits of C++ as an object oriented programming language
3. Understand how to produce object-oriented software using C++
4. Understand how to apply the major object-oriented concepts to implement object oriented programs in C++, encapsulation, inheritance and polymorphism
5. Understand advanced features of C++ specifically stream I/O, templates and operator overloading

Course Outline

Topic	LH	PH
Introduction	2	2
Elementary Programming	4	2
Data and Expressions	4	2
Selections	4	2
Loops	4	2
Methods	4	2
Arrays	4	2
Classes and Objects	4	4
Strings and Text I/O	4	2
Case study, Project Presentation &demo	2	2
Total	36	24
Total CH	60	

Mode of Delivery

- Lecture: 36 hours (24 hours for lectures and 12 hours for tutorials)
- Practicals: 24 PH

Mode of Assessment

Progressive Assessment and exams:

- | | |
|--------------------|-----|
| 1. Tests/Quizzes | 20% |
| 2. Labs/Assignment | 10% |
| 3. Course Project | 20% |
| 4. Final Exam | 50% |

Basic Reading List

1. Lewis, John & Loftus, William. **Java Software Solutions: Foundations of Program Design, 7th ed.** Addison-Wesley, 2012. ISBN 0-32137337-5.
2. Deitel, H. & Deitel, P.J. **Java How to Program 7th ed.** Prentice Hall, 2007. ISBN 978-0132222204.
3. Y. Daniel Liang, **Introduction to Java Programming 9th ed.** Pearson Hall, 2013. ISBN-13: 978-0-13-213080-6.

Course Name	Data Structures and algorithms
Code	CSC1203
Type	Core

Prerequisite	None
Level	Year 1 Semester 2

Course Description

The course is intended to provide the student with an ability to design and discover fast algorithms, improve existing algorithms and rationally choose between algorithms in order to solve a problem. The programming language to be used is Java for all implementations of the data structures and algorithms. Besides introducing the basic language (Java) and tools for algorithm analysis, the course will also cover several specific problems and general design paradigms.

Course Objectives

The objectives of this course will be to:

- a) Provide a solid background in algorithms for computer science students
- b) Assess how the choice of data structures and algorithm design methods impacts the performance of programs.
- c) Choose the appropriate data structure and algorithm design method for a specified application.
- d) Solve problems using data structures such as linear lists, stacks, queues, hash tables, binary trees, heaps, tournament trees, binary search trees, and graphs and writing programs for these solutions.

- e) Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions.

Learning Outcomes:

By the end of the course, students are expected to:

- Be able to design, implement and discover good algorithms
- Be able to improve a given algorithm
- Be able to mathematically analyse the efficiency of an algorithm
- Be able to choose rationally between two algorithms for the same task
- Be competent with Java programming language.

Course Outline

Topic	LH		PH
Fundamentals - Basic Programming Model - Data Abstraction - Bags, Queues, and Stacks - Algorithms - The Growth of Functions - Complexity of Algorithms - Analysis of Algorithms - Design of Algorithms	8	4	8
Sorting - Elementary Sorts - Merge sort - Quicksort - Priority Queues - Applications of sorting - Sorting algorithms	4	2	4
Searching - Symbol Tables - Binary Search Trees - Balanced Search Trees - Hash Tables - Applications of searching - Search algorithms	4	2	4
Graphs	4	2	4

– Undirected Graphs – Directed Graphs – Minimum Spanning Trees – Shortest Paths – Graph algorithms			
Strings – String Sorts – Tries – Substring Search – Regular Expressions – Data Compression			
	4	2	4
Total	24	12	24
Total	60		

Mode of Delivery

- Lectures: 36 hours (Include tutorials)
- Practical: 24 hours

Mode of Assessment

- Tests and assignments 40%
- Final examination 60%

Basic Reading List

- Mark Allen Weiss, *Data Structures and Algorithms Analysis in Java*, Pearson Education, 3rd edition, 2011.
- Robert Sedgewick and Kevin Wayne, *Algorithms*, 4th Edition, 2011.
- Stuart Reges and Marty Stepp, *Building Java Programs: A Back to Basics Approach* (3rd Edition), 2015.
- E. Horowitz, S. Sahni and S. Rajasekaran. *Fundamentals of Computer Algorithms*

Course Name **Database Systems**
Code **CSC1204**

Prerequisite None
Level Year 1 Semester 2

Course Description

Database Systems covers database design and the use of databases in applications, with a short introduction to the internals of relational database engines. The course leads to the understanding of how database systems are designed, implemented and maintained which is very essential to an Information technology/Computer Science student since databases are now the underlying framework of information systems, and have fundamentally changed the way many organizations operate. This course focuses on equipping the student with the knowledge and these skills.

The objective of Database Systems is to provide comprehensive knowledge of design, structure, theory and usage of a simple database management system. Students will learn the fundamental concepts of database management, including database design, languages and implementation techniques and develop skills to apply those concepts in practical data management applications

Learning Outcomes

By the end of this course, students will be able to;

1. Understand the database approach and environment.
2. Understand data analysis techniques and carry out data analysis and modelling.
3. Apply the principles of database design in a practical situation.
4. Design databases with minimal redundancy.
5. Understand the concepts underlying the maintenance of database consistency and concurrent transaction execution.
6. Have an in-depth understanding of how to achieve database security.

Course Outline

Topic	LH	TH	PH
Introduction to Databases – Traditional File-Based Systems, Database Approach, Roles in the Database Environment, History of Database Management Systems, Advantages and Disadvantages of DBMSs – Database Environment, Database Languages, Data Models and Conceptual Modelling, Functions of a DBMS – Database Architectures, Multi-user DBMS, Web Services and Service-Oriented, Distributed DBMSs, Data Warehousing, Cloud Computing, Components of a DBMS and Oracle Architecture	2		
The Relational Model, Integrity Constraints and Views	2		
Relational Algebra and Relational Calculus	2	2	
SQL	8	4	12

– Writing SQL Commands, Data Manipulation – The SQL Data Types, Integrity Enhancement Feature, Data Definition and Views – The SQL Programming Language – Subprograms, Stored Procedures, Functions, and Packages – Triggers – Recursion – 2011 SQL			
Entity—Relationship Modelling – Entity Types, Relationship Types, Attributes, Strong and Weak Entity Types, Attributes on Relationships, Structural Constraints, Specialization/Generalization, Aggregation, Composition, Database project	4	2	4
Object-Relational DBMSs – Storing Objects in a Relational Database, Introduction to Object-Relational Database Systems, Object-Oriented Extensions	2	2	4
Normalization – The Purpose and process of Normalization, How Normalization Supports Database Design, Data Redundancy and Update Anomalies, Functional Dependencies, First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF) – Overview of Boyce—Codd Normal Form (BCNF), Fourth Normal Form (4NF), Fifth Normal Form (5NF)	4	2	4
Total	24	12	24
Total		60	

Mode of Delivery

- Lectures: 40 LH
- Practical: 20 PH

Mode of Assessment

- Practical problem sets - 20%
- Programming project - 10%
- Tests and quizzes - 10%
- Final Examination - 60%

Basic Reading List

1. Connolly & Begg. Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition, Pearson, 2015

2. Elmasri & Navathe. Fundamentals of Database Systems, 7th Edition, Pearson, 2016
3. Silberschatz A., Korth H. F. and Sudarshan. 2002. Database Systems Concepts, 4th Edition. New York:McGraw-Hill.
4. Greg R. 2001.Principles of Database Systems with Internet and Java Application. USA: Addison Wesley Publishing Company.

Course Name	Applied Linear Algebra
Code	MT2105

Level	Year 2 Semester 1
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Course Description

The purpose of this course is to use a computational approach to present an introduction to basic concepts of linear algebra. By the end of this course, a successful student will be able to solve and analyse linear equations, use matrices to solve problems, perform matrix algebra, analyse properties of sets of vectors and vector spaces, analyse linear transformations and apply them in geometric settings, compute eigenvalues and eigenvectors and use them to solve problems, and apply inner products to examine orthogonality and its application to the method of least squares. Emphasis will be placed on the learning and understanding of definitions and abstractions in mathematics, as well as the study of the use of linear algebra in real-world problems. A more detailed list of topics is given later under Tentative Course Schedule.

Learning Outcomes

After completing this course, students will be able to

1. Solve linear systems using matrices and Gaussian elimination, understand the different types of solutions that are possible, and use these ideas in applied problems.
2. Perform the common operations of matrix algebra and use them to solve applied problems.
3. Compute the determinant of a square matrix and understand its properties.
4. Understand the ideas of linear independence, spanning set, basis, change of basis of a linear transformation, rank of a matrix, vector space, subspace, and their application to applied problems.
5. Understand eigenvectors and eigenvalues, how they characterize the action of some linear transformations, and how to use them to solve applied problems.

Course Content

Topic	LH	TH	PH
matrix operations	1		
systems of linear equations	2		
inverse of a matrix	1		
partitioned matrices	1		
properties of determinants	1		
vector spaces, subspaces	2	1	
linear combinations, span	2		
null space, column space	2		
linear independence, bases	2		
linear transformations	1	1	
computer graphics	1	1	1
dimension of a vector space, rank, isomorphisms	2	1	
coordinate systems, change of basis	2		
MATLAB tutorial, orthogonality	2	1	1
orthogonal sets, orthogonal projections	2		
Gram-Schmidt orthogonalization	1		
QR factorization	1		
least squares, linear models	2		
eigenvalues and eigenvectors	2		
diagonalization	1		
singular value decomposition	1		
vector norm, matrix norm, condition number	2		
linear programming	2	1	1
Total hours	36	6	3
TOTAL CH	45		

Mode of delivery

- Lecture hours
- Tutorial hours
- Practical hours

Mode of Assessment

- Four Progressive Assessment: (combined must equal 40%)
- Final Exam 60%

Textbooks

1. Linear Algebra and Its Applications, 5th edition, by David C. Lay, Steven R. Lay, Judi J. McDonald, Pearson, 2016, ISBN-13: 978-0321982384 ISBN-10: 032198238X

Course Name	Integral Calculus
Code	MTH1202

Level	Year 1 Semester 2
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Course Description

Calculus is a transition course to upper-division mathematics and computer science courses. Students will extend their experience with functions as they study the fundamental concepts of calculus: limiting behaviours, difference quotients and the derivative, Riemann sums and the definite integral, antiderivatives and indefinite integrals, and the Fundamental Theorem of Calculus. Students review and extend their knowledge of trigonometry and basic analytic geometry. Important objectives of the calculus sequence are to develop and strengthen the students' problem-solving skills and to teach them to read, write, speak, and think in the language of mathematics. In particular, students learn how to apply the tools of calculus to a variety of problem situations.

Course Objectives

By the end of this course, students should be able to

- a) Understand concepts rather than merely mimic techniques
- b) Demonstrate understanding by explaining in written or oral form the meanings and important applications of concepts
- c) Construct and analyze mathematical models of real-world phenomena, including both discrete and continuous models
- d) Distinguish between discrete and continuous models, and make judgments about the appropriateness of the choice for a given problem
- e) Understand the relationship between a process and the corresponding inverse process

Learning Outcomes/ Goals

- a) Demonstrate competency in the areas that comprise the core of the mathematics major.

- b) Demonstrate the ability to understand and write mathematical proofs.
- c) Be able to use appropriate technologies to solve mathematical problems.
- d) Be able to construct appropriate mathematical models to solve a variety of practical problems.
- e) Obtain a full-time position in a related field or placement in graduate school within one year of graduation.

Course Outline

Topic	LH	TH
Real and complex numbers	2	
Functions	2	1
Sequences	2	
Series	2	1
Limit of a function at a point	2	
Continuous functions	2	1
The derivative	4	
Integrals	4	1
Definite integral	2	
Applications of integrals	4	1
Improper integrals	2	
Wallis' and Stirling's formulas	2	2
Numerical integration	2	
Differential equations	4	2
Total Hours	36	9
Total CH	45	

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.
- Interactive, communicative, discussion based learning and task based learning.

Mode of Assessment

Assessment	mark
Course works	
• Exercises	5%
• Quizzes	10%
• Tests	25%
Final exams	60%

Basic Reading List

1. Kenneth Kunen. Set Theory. Studies in Logic and the Foundations of Mathematics, Vol. 102, North-Holland, Amsterdam, 1980.
2. Edmund Landau. Grundlagen der Analysis, 4th ed. American Mathematical Society, New York, 1965.
3. Wolfgang Walter. Analysis 2, 5th ed. Springer-Verlag, Berlin, 2002 (German).

Course Name **Political Economy of Africa and Development**
Code DVS 1203

Prerequisite None
Level Year 1 Semester 2

Prerequisite: None

Purpose/Aim: The aim of this course is to increase students' capacity to use their growing knowledge of political economy of development in rigorous analysis of a subset of critical issues facing policy makers in sub-Saharan Africa

Description: The aim of this course is to increase students' capacity to use their growing knowledge of political economy of development in rigorous analysis of a subset of critical issues facing policy makers in sub-Saharan Africa. The African continent stands high among the many countries facing development challenges. The exposure to which Africa received from Europe during colonialism; the post-colonial situation and the current globalization drive leave Africa in a development dilemma. African development has gained prominence in the development discourse. Every development studies service course student that is interested in understanding the mechanisms and processes through which social transformation can be created needs to understand the position and processes through which Africa has transformed. The course interrogates African development problem taking into consideration the forces that shaped the colonization of Africa; colonialism and its social, economic and political impact on Africa

Objectives:

1. Understand development challenges that face Africa today from the historical and contemporary perspectives
2. Analyze the historical processes that have shaped the development of Africa
3. Suggest viable recommendations for Africa's development challenges from a well-

informed perspective.

Learning outcomes

At the end of the course, students will be expected to:

- Know development challenges that face Africa today from the historical and contemporary perspectives
- Suggest viable recommendations for Africa's development challenges from a well-informed perspective
- Apply their knowledge in constructing a critical analysis of the issues for a particular African country

Course content:

Pre-colonial, Colonial and Postcolonial Africa

- Pre-colonialism- Kingdoms, socio-economic arrangements & classes, pre-colonial economy 6 Hours
- Slave Trade 4 Hours
- Colonialism- Actors, colonial economy, production- colonial economies-settler economy, plantation economy and peasantry economies, Regional Division of labor 6 Hours
- Resistance to Colonialism in Africa: resistors and Collaborators, Passive and Active Resistances (Mau Mau, Nama-Herero, MajiMaji etc) 6 Hours
- National Movements (and personalities) and Fight for Independence 3 Hours

Regionalism and African Integration

6 Hours

OAU/AU, SADC, ECOWAS, COMESA, EAC

Development and Underdevelopment

- Theories of Development 2 Hours
- Approaches to development: Top-Bottom, Bottom-up approaches 2 Hours
- Causes/factors and actors of underdevelopment 2 Hours

- Poverty: Poverty analysis, causes, dimensions, way forwards 2 Hours

Democracy, good governance, human rights and Development 6 Hours

Mode of delivery:

- Lectures
- Case studies and class exercises.
- Group assignments and presentations.
- Group reading.

Mode of Assessment:

No	Mode of Assessment	Mark (%)
1	Mid-semester exam or take home assignment	30
2	Class attendance and Students presentations	10
3	Final Examinations	60
	TOTAL	100

Instructional Materials

5. LED Projectors
6. Laptops
7. Teaching boards and writers
8. Rooms with enough chairs and tables.

Reading materials:

1. Rodney Walter. (1972). *How Europe underdeveloped Africa*. London: Zed Books
2. Daron Acemoglu and James Robinson (2012). *Why Nations Fail: The Origins of Power, Prosperity and Poverty*
3. Boahen, A. A. (1987). *African Perspective on Colonialism*, Baltimore, Maryland
4. Mamdani, Mahmood. (1996). *Citizens and Subject: Contemporary Africa and the Legacy of Late Colonialism*. Princeton, NJ: Princeton University Press
5. Omotola, J. Shola&Saliu, Hassan. (2009). "Foreign aid, debt relief and Africa's development: problems and prospects," *South African Journal of International Affairs*, Vol. 16, Issue 1, pp. 87- 102

6. Moyo, Dambisa. (2009). *Dead Aid: Why aid is not working and how there is a better way for Africa*, New York: Farrar, Straus & Giroux
7. Nkrumah, K. 1965, *Neo-colonialism: the last stage of Imperialism*, London: Heinemann
8. Nyerere, J.K. 1997, 'Africa Must Unite', edited excerpts from a public lecture delivered in Accra to mark Ghana's fortieth Independence Day anniversary celebrations, United New Africa Global Network website:
<http://www.unitednewafrica.com/Africa%20Unite.htm>
9. Museveni, Y. K (1992) What is Africa's Problem? Kampala: NRM Publishers
10. Kothari Uma and Minogue Martin (Eds), (2002); *Development theory and practice*, Palgrave, London.
11. Fanon, Frantz. *The Wretched of the Earth*. Trans. Constance Farrington. New York: Grove, 1966.
12. Chazaret el (1999), Politics and Society in Contemporary Africa, 3rd Edition, Lynne Reinner publishers.
13. Ocheni Stephen & Nwankwo Basil C. (2012). Analysis of Colonialism and Its Impact in Africa. Canadian Academy of Oriental and Occidental Culture, Vol. 8, No. 3, 2012, pp. 46-54
14. Ayittey B. N. George. (1999), Africa in Chaos, St. Martinis, Griffin, New York
15. Iliffe, John. "The Organization of the Maji Maji Rebellion," *Journal of African History*, 8:3 1967, pp 495-512
16. Kanogo, Tabitha. *Squatters and the Roots of Mau Mau*. Athens: Ohio University Press, 1987.
17. Amartya Sen (1999) *Development as freedom*, Oxford: OUP
18. Willis Katie. (2005). *Theories and Practices of Development*. London: Routledge – Taylor and Francis

Course Name	Networks Certification (CCNA)
Code	CSC1206

Prerequisite	None
Level	Year 1 Recess Term

Course Description

The Cisco Certified Network Associate (CCNA) Routing and Switching is a gateway to entry-level networking jobs and IT careers. The curriculum consists of 4, 70-hour courses: Introduction to Networks, Routing and Switching Essentials, Scaling Networks, and Connecting Networks. The first 2 courses prepare you for the Cisco CCENT certification exam or to study CCNA Security. All 4 courses are recommended before taking the Cisco CCNA Routing and Switching certification exam.

Course Objectives

- a) Develop a working knowledge of routing, switching, network applications, protocols, and services.
- b) Study with an instructor in the classroom and access expert content online anytime.
- c) Practice what you learn on both real equipment and Cisco Packet Tracer, a network configuration simulation tool.
- d) Get immediate feedback on your work through built-in quizzes and tests.
- e) Prepare for the workplace with collaborative projects and presentations.
- f) Connect with the global Cisco Networking Academy community.

Curriculum Features

The Networking Academy curriculum is delivered online using a proprietary learning platform. Cisco provides the curriculum, learning platform, and support resources. The department hire instructors, provide equipment, and enrol students. The curriculum provides:

- a) An engaging, up-to-date curriculum created by experts in technology and instructional design
- b) Hands-on labs and online activities for the practical application of learned skills
- c) Cisco Packet Tracer, a network configuration simulation tool, used for teaching, gaming, and team building
- d) Instructor guides, community forums, and professional development support instructors
- e) Innovative online assessments include embedded activities, simulations, text questions, and performance-based reviews

4.3 Year 2 Semester 1

Course Name	Operating Systems
Code	CSC2101

Prerequisite	None
Level	Year 2 Semester 1

Course Description

This is an undergraduate course that provides a thorough introduction into the topic of operating systems. The students will be expected to have programming and reasoning abilities, as well as background on data structures and computer organization. This course is devoted to a structured survey of OS concepts and practices. Similarly, certain prominent OS will as well be introduced.

Course Objectives

Upon successful completion of this course, the student will be able to:

1. Understand computer system structures.
2. Understand process management in operating systems, including such topics as process definition, threads, scheduling, synchronization and deadlocks.
3. Understand memory management in operating systems, including such topics as main and virtual memories, memory allocation and paging and segmentation.
4. Understand storage management in operating systems, including such topics as file-system interface, mass storage structure and I/O systems.
5. Understand distributed systems, including such topics as network-based operating systems, distributed file systems and distributed coordination.

Learning Outcomes:

1. Explain the objectives and functions of modern operating systems.
2. Describe how operating systems have evolved over time from primitive batch systems to sophisticated multiuser systems.
3. Analyse the trade-offs inherent in operating system design.
4. Understand how Jobs are scheduled from memory for execution
5. Understand Memory management
6. Describe various memory page replacement algorithms

Course Content

Topics	LH	TH	PH
Introduction to Operating systems	4		
Operating system Structure	2	2	
Process and threads – Process and programs, concurrency, process states, thread of control, interacting processes.	6	2	2
Scheduling – Non pre-emptive and pre-emptive scheduling policies, scheduling in practice, real-time scheduling, examples of scheduling in UNIX, Linux and Windows	6	2	2
Deadlocks	2	2	2
Memory Management – Main memory - Memory hierarchy, address spaces, static and dynamic memory, memory allocation to process, continuous and non-continuous memory allocation, swapping and relocation, paging, segmentation. – Virtual Memory	4	4	2
File System and Input/output Control System (IOCS) – Files, directories, pathnames and filenames, multiple file systems, file types, file sharing, links and shortcuts, file locking, file attributes, disk structure, examples of UNIX, Linux and Windows file systems. – Architecture of the IOCS, device drivers, types of devices, buffering, device driver structure.	4	2	2
Multiprocessor systems – Clients and servers, distributed file systems, distributed processing, introduction to thin client computing, multicomputer systems	4	2	2
Total	32	16	12
Total	60		

Mode of Delivery

- Lecture hours
- Tutorials hours
- Practical hours

Mode of Assessment

- Coursework (40%)- includes Tests and class exercises

- Final Examination (60%)

References

1. "Operating System Concepts" 9th Edition by Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Publisher: Wiley; 9th edition, 2012, ISBN-10: 1118063333 ISBN-13: 978-1118063330.

Course Name	Object Oriented Programming II
Code	CS2102

Prerequisite	None
Level	Year 2 Semester 1

Course Description

The course covers the use of classes and the fundamentals of object-oriented design, including: class composition, Java basics, inheritance, polymorphism, exception handling and unified modelling language (UML).

Course Objectives

At the end of this course, the student should be able to:

1. Develop knowledge of the object-oriented language (Java) and analysis and design (UML).
2. Explain how to solve problems, transform the solution into a program written in the java programming language.
3. Demonstrate how to test, verify, and debug object-oriented programs; and
4. Create programs using object – oriented principles.

Learning Outcome

1. Explain the benefits of object oriented design and understand when it is an appropriate methodology to use.
2. Understand how to produce object-oriented software using Java
3. Understand how to apply the major object-oriented concepts to implement object oriented programs in Java, encapsulation, inheritance and polymorphism
4. Design object oriented solutions for small systems involving multiple objects

Course Outline

Topic	LH	PH
Object-Oriented Concepts and UML	2	4
Inheritance	2	4
Encapsulation	2	4
Method Overriding	2	4
Polymorphism	2	4
Abstract Class	2	4
Interfaces	2	4
Exceptions	2	4
Java Web	2	4
JDBC	2	4
	20	40
Total CH	60	

Mode of Delivery

- Lecture hours: 22
- Practical hours: 48

Assessment Criteria

Progressive Assessment and exams:

- | | |
|------------------------|------|
| 1. Tests/Quizzes | 10% |
| 2. Problem sets | 15%% |
| 3. Programming project | 15% |
| 4. Final Exam | 60% |

Basic Reading List

1. Introduction to Programming in Java: An Interdisciplinary Approach by Sedgewick, Robert
2. Building Java Programs. A Back to Basics Approach 2011 by Stuart Reges
3. Introduction to Java Programming by Daniel Liang (10th Edition)
4. Lewis, John & Loftus, William. Java Software Solutions: Foundations of Program Design, 7th edition.
5. Simon Bennett, Steve McRobb & Ray Farmer. Object-Oriented Systems Analysis and Design Using UML, 3rd edition.
6. Danny Poo, Derek Kiong & Swarnalatha Ashok. Object-Oriented Programming and Java, 2nd ed. Springer, 2008. ISBN-13: 978-1846289620.

Course Name	Web Programming
Code	CSC2103

Prerequisite	None
Level	Year 2 Semester 1

Course Description

Web programming is one of the most exciting units you can study at the university. Internet Programming examines programming models on both client and server sides. A web client is usually a browser, but may also be a custom application. The course introduces the practices of computer programming using languages common on the Internet, including HTML, XML, JavaScript, PHP, and, primarily, Java. Standard object-oriented design patterns and common Java-based libraries, distributed computing Java technologies, and multi-tier software system architecture using Java technology will be introduced. Emphasis will be on learning the technical skills and knowledge necessary for the design and development of Web-based systems.

Course Objectives

At the end of this course, students will:

1. Have a general understanding of the fundamentals of the Internet, WWW, browsers, TCP/IP, telnet, ftp etc.
2. Have basic knowledge and experience with the major web technologies. Existing and future technologies
3. Have insight into what constitutes a well-designed, usable web application
4. Know how to manage a project
5. Be able to solve various internet programming problems and be innovative thinkers
6. Be able to apply the technologies with real business

Learning Outcomes

1. Ability to implement an appropriate planning strategy for developing websites.
2. Ability to produce functional, flexible, & versatile websites.
3. Ability to locate, evaluate, & critically assess current & emerging technologies for developing websites.
4. Possess a good working knowledge of HTML5 & CSS.
5. Experience creating various small website projects.
6. An awareness of the process in creating a website & the various roles needed in that process.

Course Outline

Topic	LH	PH
JavaScript and HTML Documents – The JavaScript Execution Environment, The Document Object Model, Element Access in JavaScript, Events and Event Handling, Handling Events from Body Elements, Handling Events from Button Elements, Handling Events from Text Box and Password Elements, The DOM 2 Event Model, The canvas Element, The navigator Object, DOM Tree Traversal and Modification	2	8
Dynamic Documents with JavaScript – Positioning Elements, Moving Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Stacking Elements, Locating the Mouse Cursor, Reacting to a Mouse Click, Slow Movement of Elements, Dragging and Dropping Elements	2	8
XML – Uses of XML, The Syntax of XML, XML Document Structure, Namespaces, XML Schemas, Displaying Raw XML Documents, Displaying XML Documents with CSS, XSLT Style Sheets, XML Processors, Web Services	4	8
PHP – Overview of PHP, General Syntactic Characteristics, Primitives, Operations, and Expressions, Output, Control Statements, Arrays, Functions, Pattern Matching, Form Handling, Cookies, Session Tracking, Object-oriented PHP	8	8
Ajax	4	8

Topic	LH	PH
– Overview of Ajax, The Basics of Ajax, Return Document Forms, Ajax Toolkits, Security and Ajax		
Java Web Software – Introduction to Servlets, Storing Information on Clients, JavaServer Pages, JavaBeans, Model-View-Controller Application Architecture, JavaServer Faces	4	8
Total	24	48
Total	60	

Mode of Delivery

- Lecture hours: 20
- Practical hours: 40

Mode of Assessment

- Tests & Quizzes 10%
- Problem Sets 10%
- Course Project 20%
- Final Exam 60%

Basic Reading List

1. Robert W. Sebesta, *Programming the World Wide Web*, 8th edition, Pearson, 2015.
2. Harvey M. Deitel & Paul J. Deitel, [Internet & World Wide Web: How to Program](#), 5th Edition, 2012.
3. Simon Stobart & David Parsons, Dynamic Web Application Development Using PHP and MySQL, ISBN 9781844807536
4. Craig D. Knuckles, David S. Yuen, Web Applications: Concepts & Real World Design, John Wiley & Sons, ISBN 0-471-20458-7 (paperback) or 0-471-42929-5 (paperback, Wiley international edition).

Course Name **Database Programming**
Code **CSC 2104**

Prerequisite None
Level Year 2 Semester 1

Course Description

This subject teaches students how to design, develop and evaluate database programming and administration solutions to meet pre-defined quality characteristics of functionality

(suitability, security), usability (operability), efficiency (time behaviour, resource utilization), and maintainability (changeability, testability). Database programming and administration solutions are implemented using Oracle 11G, SQL*Plus and PL/SQL. Concepts, theories and technologies underlying the methods and techniques are introduced and explained as required. Students apply all that they have learnt to develop a small application to solve a database problem

Course Objectives

Upon successful completion of this subject, students should be able to:

1. Describe the benefits of database programming
2. Understand programming constructs available in PL/SQL.
3. Understand how triggers, stored procedure and stored functions and database packages operate.
4. Describe the use of supplied packages in the database.
5. Have basic understanding of the roles and responsibilities of a database administrator.

Learning Outcomes/ Goals

1. Design and model relational databases.
2. Document database structures and rules.
3. Maintain and retrieve data.
4. Perform basic administrative functions.
5. Perform security Administration to protect data integrity.
6. Develop databases with a variety of current industry software applications, such as Microsoft Access and SQL Server, on a variety of operating system platforms like Windows 7 and Server 2012.

Course Outline

Topic	LH	PH
Data Definition Language (DDL)	2	4
Working with Queries – Retrieving Data	2	4
Modifying data in a Table	2	4
Using SQL Functions in Queries	2	4
Working with Multi-Table Queries (JOINS)	2	4
Implementing Data Integrity	2	4
Using Databases for Software Applications	2	4
Advanced SQL – • Procedures,	2	4

• Functions, • Triggers and views		
ODBC and JDBC with Java	2	4
Intro to Big Data and Semantic web	2	4
NoSQL and Non-SQL databases and applications	2	4
Geo-Spatial, RDF and Graphing databases	2	4
Total	24	48
Total	60	

Mode of Delivery

- Formal and interactive lectures and practical problem sets
- Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

Assessment mark

Course works

- Problem sets 20%
- Quizzes and test 5%
- Course project 15%

Final exams 60%

Basic Reading List

1. Oracle PL/SQL Programming (5th Edition) ISBN 978-0-596-51446-4 by Steven Feuerstein with Bill Pribyl, Published by O'Reilly and Associates Inc.
2. "MySQL Workbench: Data Modeling & Development" by Michael McLaughlin.

COURSE NAME: Systems Analysis and Design

COURSE CODE: CSC 2105

COURSE LEVEL: Year Two, Semester One

Course Description

The course material encompasses the concepts, tools and techniques required to analyze and design business information system. This course will give an overview of Information systems, the system development methodologies and techniques, system analysis and design as system development phases.

The student will become familiar with the principle, terminology and techniques of system analysis and design of business information systems and develop critical thinking and problem solving skills. They will also improve written and verbal communication skills as required of system analysts

2. Course Objectives

Upon completion of the course the student should be able to;

- i. Understand composition of an information system
- ii. Understand system planning and feasibility study
- iii. Understand system analysis of a system
- iv. Understand different designs of a new system
- v. Understand system documentation mechanisms

3. Learning Outcomes

Upon completion of the course the student;

- i. Should know what makes up an information system
- ii. Should be able to analyse a system, propose new system and design a new system
- iii. Should be able to model current and proposed system
- iv. Should be able to design a system
- v. Should be able to elicit requirements of a system and document them prior to design

4. Methods of delivery

- ◆ Lectures
- ◆ Group discussions
- ◆ Assignments

5. Mode of Assessment

No	Assessment	Marks %
1.	Coursework	40
2.	End of Semester Examination	60

	Final Mark	100
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6. Course Outline

Topic	Sub-Topics	Contact Hours
Introduction	- Basic Concepts of Analysis and Design - information system Overview	3
The system Development Environment	- The system development Life Cycle - Different Development Methodologies	3
System Planning	Preliminary investigation and Feasibility study	4
Requirements Determination	Traditional and modern methods	4
Structuring System Requirements	- Process modeling-Data - Flow Diagrams	6
Logical modeling	- structured English - Decision Tables	6
System Design	- design strategy - Architectural Design - Interface Design - Program Design	4 3 6 6
Total Contact Hours		45

7. Reference List

1. Alan Dennis and Barbara Haley Wixom. 2003. Systems Analysis and Design, second Edition. Von Huffman Press Inc.
2. Valacich, George & Hoffer. 2000. Modern Systems Analysis and Design, fourth Edition.
3. Valacich, George & Hoffer. 2002. Essentials of System Analysis and Design, second Edition.

Course Name **Scientific thinking**

Code CSC2106

Type Core

Prerequisite None

Level Year 2 Semester 1

Course Description

The course emphasizes the unifying aspects of the scientific approach to the study of nature and human behavior. About one-third of the course is devoted to scientific inquiry and investigation. The course focuses on fact identification and concept formation and testing. In the remaining parts, the students are exposed to applications of the approach in various disciplines. The course sets some basic concepts and theories of science into broad historical, philosophical, and cultural context and traces the development of these theories to their present status. This serves the double purpose of acquainting students with the appropriate setting in which a given idea gained relevance and exposing them to the evolution toward the current methods of investigation. Moral and ethical issues in science are examined.

Course Objectives

The objectives of the course is to have students learn to define science use reasoning skills such as, analysis, synthesis, including, deducing, increasing, apply the methods science to solve problems, use creative thinking skills in real situations, and use research methods in engineering sciences.

Learning Outcomes

At the end of the course, a student should be able to;

1. Select, analyze, interpret and evaluate a range of source materials.
2. Describe patterns or relationships in large amounts of written and/or visual information.
3. Develop a well-supported, clearly articulated argument to support a view and use it to justify one or more conclusions.
4. Evaluate available written and/or visual information, evidence and argument for reliability and authority/usefulness (e.g.; observation, testimony, measurement, experiment).

Methods of Delivery

1. Lectures
2. Tutorials

3. Practicals

Mode of Assessment

Assessment	mark
Course works	
• Problem sets	20%
• Quizzes and test	5%
• Presentation and discussion	15%
Final exams	60%

Detailed Content

Topics	Sub Topics	Contact Hours
Overview	Thinking Patterns Development	03
Meaning & Construction of Science	- Introduction - Scientific Values & Directions	04
Science and non-science	- Introduction - Science, Engineering & Technology. - Properties of science - Objectives of science - Postulates of scientific thinking	08
Mental operations used in science	- Introduction - Scientific guessing - Types of deductions - Presentation	08
Research methods in mathematical sciences	- Introduction - Postulates, definitions - Research methods in natural Sciences	06
Experiments & observations	- Scientific postulates & their conditions - Verification of scientific postulates.	05
Problem solving	- Introduction - General methods of problems solving	05
Creative thinking	- Introduction - Fluency types. - Flexibility & originality - Basics of brain storming	06
Total Contact Hours		45

References

1. S. Bloom, M. Engelhart, E. Furst, W. Hill, and D. Krathwohl. Taxonomy of Educational Objectives. David McKay Company, Inc, NY, 1956.
2. S. Choy and P. Cheah. Teacher Perceptions of Critical Thinking Among Students and its Influence on Higher Education, volume 20(2). 2008.
3. B. Fagin, J. Harper, L. Baird, S. Hadfield, and R. Sward. Critical thinking and computer science: implicit and explicit connections, volume 21(4).
4. T. Moore. The critical thinking debate: how general are general thinking skills?, volume 23(1).
5. P. Sharma and M. Hannafin. Scaffolding critical thinking in an online course: An exploratory study, volume 31(2). 2004.
6. [6] J. Tapper. Student perceptions of how critical thinking is embedded in a degree program, volume 23(2).

Course Name **Differential Equations**

Code **MTH2101**

Type **Core**

Prerequisite **None**

Level **Year 2 Semester 1**

Course Description

The construction of mathematical models to address real-world problems has been one of the most important aspects of each of the branches of science. It is often the case that these mathematical models are formulated in terms of equations involving functions as well as their derivatives. Such equations are called differential equations. If only one independent variable is involved, often time, the equations are called ordinary differential equations. The course will demonstrate the usefulness of ordinary differential equations for modeling physical and other phenomena. Complementary mathematical approaches for their solution will be presented, including analytical methods, graphical analysis and numerical techniques. The basic content of the course includes;

- i. first order equations
- ii. mathematical models
- iii. linear equations of second order
- iv. the Laplace transform

- v. linear systems of arbitrary order and matrices
- vi. nonlinear systems and phase plane analysis
- vii. numerical methods

Course Objective

Upon completion of the course students should be able to:

1. Analyze real world scenarios to recognize when ordinary differential equations (ODEs) or systems of ODEs are appropriate, formulate problems about the scenarios, creatively model these scenarios (using technology, if appropriate) in order to solve the problems using multiple approaches, judge if the results are reasonable, and then interpret and clearly communicate the results.
2. Recognize ODEs and system of ODEs concepts that are encountered in the real world, understand and be able to communicate the underlying mathematics involved to help another person gain insight into the situation.
3. Work with ODEs and systems of ODEs in various situations and use correct mathematical terminology, notation, and symbolic processes in order to engage in work, study, and conversation on topics involving ODEs and systems of ODEs with colleagues in the field of mathematics, science or engineering.

Learning Outcomes

At the end of the course, a student should be able to;

1. Demonstrate an understanding of the theory of ODEs and their applications in:
 - a) At least two proctored exams, one of which is a comprehensive final that is worth at least 25% of the overall grade; proctored exams should be worth at least 50% of the overall grade
 - b) and at least one of the following:
 1. Take home exam(s)
 2. Quizzes
 3. Computer lab assignments
 4. Homework
 5. ODE related Community-Based Learning
2. Demonstrate the ability to communicate with colleagues on the topics of ODEs and systems of ODEs in:
 - a) At least one group or individual project with written report and/or oral in-class presentation.
 - b) and at least one of the following:
 - i. Participation in in-class discussions.
 - ii. In-class group activities
 - iii. Attendance

Course Outline

Topic	LH	TH
Chapter 1 - Differential Equation Models.	4	2
Chapters 2, 3, 6 - First-Order Equations and Applications:	4	2
Chapter 7 - Linear Algebra and Linear Systems of Equations:	4	2
Chapters 8, 9 - Systems of Differential Equations:	4	2
Chapter 4 - Second-Order Linear Equations:	4	2
Chapter 5 - The Laplace transform:	4	4
Chapter 10 - Nonlinear systems:	4	3
Total Hours	28	17
Total CH	45	

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.

Mode of Assessment

Assessment	mark
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Course works	
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- | | |
|-------------|-----|
| • Exercises | 5% |
| • Quizzes | 10% |
| • Tests | 25% |

Final exams	60%
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Basic Reading List

1. Kreyszig, E. (2011), Advanced engineering mathematics, 10th edn, Wiley.

COURSE NAME: Political Economy of Uganda

CODE: DVS 2105

COURSE LEVEL: Year 2 Semester 1

Prerequisite: Introduction to Political Economy of Uganda

Purpose/Aim: The aim of the course is to present and stimulate debate on the ways in which Uganda's historical development has influenced existing institutions, political structures and processes, society and culture, and to emphasize opportunities and potential risks to future development.

Description: This development studies course is designed to provide a background for a detailed analysis of processes of political and social change in Uganda, and their implications for the country's social, political and economic development. The aim of the course is to present and stimulate debate on the ways in which Uganda's historical development has influenced existing institutions, political structures and processes, society and culture, and to emphasize opportunities and potential risks to future development. This course gives an overview of Uganda's political economy from the colonial era through the years of economic reforms. This course touches on the various development plans for the country. There is a study of the various economic policies of the various regimes.

Objectives: To enable students understand the factors that have delayed Uganda's development.

1. To Identify specific policies of the various regimes
2. Give a general overview of the political economy of Uganda since independence.
3. To analyze Uganda's historical and current political and economic perspectives and how these have affected the development of this nation.
4. To identify the root causes of the development problems and possible interventions
5. To foster a spirit of nationalism and also ensure that students merge out as good ambassadors of good governance.

Learning outcomes

At the end of the course, students will be expected to:

1. Comprehend the factors that have delayed Uganda's development.
2. Identify specific policies of the various regimes
3. Give a general overview of the political economy of Uganda since independence.
4. Analyse Uganda's historical and current political and economic perspectives and how these have affected the development of this nation.

5. Identify the stem of the development problems and what can be done to alleviate them

Course content:	Hours
• Pre-colonial Ugandan Society-Production, politics, kingdoms	4
• Colonialism-Colonial economy, colonial regional division of labor,	4
• Responses to colonialism	3
• Nationalism and the birth of Political Parties	3
• Economic Reforms after Independence	2
• Import Substitution	1
• Industrial Policies	1
• Reforms in the 1970s	1
• Amin's Economic War	1
• 1980s World Bank Reforms I and II	1
• Structural Adjustment Programs Before and After 1986	1
 National Resistance Movement	
• The 10/15 points programme	1
• Political and Economic Reforms during NRM: 1986-2000- Privatization, Economic Liberalization, The 1995 Constitution	Decentralization, 3
• The Local Government Act: Gender Inequality and Equality Issues (Affirmative Action)	1
• AGOA	1
• Rural Development Policies and Strategies	1
• Plan for Modernization of Agriculture in Uganda: impact of PMA, Challenges	Successes and 2
• Uganda's role in Regional and African Context	3

- Peace Keeping missions in Somalia, DRC etc 3
- Ideology, Patriotism and Nationalism in Uganda 6

No.	COURSE OUTLINE	LH	PH	TH	CH
8.	Pre-Colonial Uganda	6	0	3	
9.	Colonial Uganda	6	0	3	
10.	Struggle for Independence	6	0	0	
11.	Post-Colonial Uganda	6	0		
12.	National Resistance Movement	9		3	
13.	Uganda's role in Regional and Global Context	6		3	
14.	Ideology, Patriotism and Nationalism in Uganda	6		3	
	Total CU	45		15	

Mode of delivery:

- Lectures
- Case studies and class exercises.
- Group assignments and presentations.
- Group reading.

Mode of Assessment:

No	Mode of Assessment	Mark (%)
1	Mid-semester exam or take home assignment	30
2	Class attendance and Students presentations	10
3	Final Examinations	60
	TOTAL	100

Instructional Materials

9. LED Projectors
10. Laptops
11. Teaching boards and writers

12. Rooms with enough chairs and tables.

Reading materials:

1. Mamdani M. (1976). *Politics and Class Formation in Uganda*. London: Monthly Review Press
2. Hansen, H & Twaddle M. (Eds). (1991). *Changing Uganda: The Dilemmas of Structural Adjustment and Revolutionary Change*. London: Villers Publications
3. Jorgensen, J. J. (1982). *Uganda: A Modern History*. Croom Helm Ltd
4. Kabwegyere, Tarsis B.(1995). *The Politics of State Formation and Destruction in Uganda*, Kampala: Fountain Publishers Ltd
5. Karugire, S. (1996). *Roots of Instability in Uganda*. Kampala: Fountain Publishers
6. Karugire S. R. (1980). *A Political History in Uganda*. Nairobi and London: Heinemann
7. Kiwanuka Semakula. (1973). *From Colonialism to Independence: A Reappraisal of Colonial Policies & African Reactions 1870 – 1960*, Nairobi: Kenya Literature Bureau
8. Moncrieffe Joy. (2004). *Uganda's Political Economy: a synthesis of major thought*
9. Villadsen, Soren and Francis Lubanga (Ed) (1996). *Democratic Decentralization in Uganda: A New Approach to Local Governance*. Kampala: Fountain Publishers, 1996

4.4 Year 2 Semester 2

Course Name	Software Engineering
Code	CSC2201

Prerequisite	None
Level	Year 2 Semester 2

Course Description

This course is an introduction to software engineering. Software engineering includes all aspects of professional software production, and is especially important for safety-critical and mission-critical software. It is also crucial for very large complicated software projects. It includes documentation, management, processes, requirements, design models,

computer programs, validation, verification, cost estimation, management, and other aspects of the development process.

The students will learn the fundamental components of software engineering, and how complex software systems are developed so as to minimize errors and maximize the usefulness of the software. They will also learn the terminology, accepted practices, and procedures used in software engineering.

Course Objectives

Students who successfully complete this course will be able to:

1. Explain the importance of safety-, mission-, business-, and security-critical systems;
2. Demonstrate knowledge of the importance of good software engineering practices for critical systems;
3. Describe and explain the terminology, accepted practices, and procedures used in software engineering;
4. Explain the differences among software engineering, computer science and systems engineering;
5. Decide which computer languages are well suited to modern critical systems (and explain why);
6. Explain a variety of life-cycle models; among others.

Course Outline and Content Description

Topic	LH	PH
Introduction to software engineering, changing nature of software Role of management in software development Mission- and Safety-Critical issues	4	0
Software lifecycle models ▪ Water fall model ▪ Incremental ▪ Evolutionary ▪ Unified process Process of selecting a particular life cycle approach	4	0
Software Requirements engineering; ▪ Feasibility ▪ Elicitation ▪ Requirements Analysis and specification ▪ Validation and documentation ▪ Management	6	0
Software Design ▪ Conceptual and technical designs ▪ Design strategy	5	3

▪ Computer Aided Software Engineering		
Software project planning and management	5	0
Software Metrics		
▪ Cost Estimation		
▪ Size estimations		
▪ Resource estimations	5	0
Software Development, ReUse, and Evolution	2	8
Verification, Validation, Testing and maintenance	2	5
Emerging Technologies	2	4
Total	35	20
Total	60	

5. Assessment Criteria

For the final grade, students will be assessed on their attendance, participation and performance in class through graded assignments like;

Continuous assessment (Class Assignments, presentations, practicals, coursework tests and continuous semester-long group project) which constitutes 40% of the final mark plus a final sit-down examination which also constitutes 60% of the final mark.

Readings Text

1. Roger S Pressman, "Software Engineering – A Practitioner's Approach", McGraw Hill, Sixth Edition 2005).
2. Ian Sommerville, "Software Engineering", Addison Wesley, Seventh Edition 2004

Course Name **Networks and Systems Administration**
Code **CSC2202**

Prerequisite None
Level Year 2 Semester 2

Course Description

Virtually all organizations have IT needs. It is the role of the IT professional to design, select, apply, deploy and manage computing systems to support the organization. This knowledge area consists of those skills and concepts that are essential to the administration of operating systems, networks, software, file systems, file servers, web systems, database systems, and system documentation, policies, and procedures.

This course addresses both the technology of computer systems and the users of the technology on an equal basis. It is about putting together a network of computers, getting them running and then keeping them running in spite of the activities of users who tend to cause the systems to fail. Topics covered include defining and controlling user access to machines, managing system storage, configuration, and control of peripherals, troubleshooting common problems, networking, and performance monitoring and management. Fundamentals of UNIX/ Linux Shell Programming shall be covered.

Learning Outcomes

Having successfully completed this course, a student should be able to: -

1. Use Windows and Linux operating systems.
2. Perform basic administration under Windows and Linux.
3. Design a network which is logical and efficient.
4. Deploy large numbers of Computers which can be easily upgraded later.
5. Demonstrate the services are needed.
6. Plan and implement adequate security.
7. Provide a comfortable environment for users.
8. Develop ways of fixing errors and problems which occur. (Independent Problem solving).
9. Understand how to use the enormous amount of knowledge which increases every year.

Learning Outcome	Student Outcomes - Graduates should be able to:
1, 2, 3, 4, 5, 6, 7, 8	a. Apply the knowledge, techniques, skills, and modern tools of the discipline to engineering technology activities
9	b. Apply knowledge of analog and digital electronics, computers, networks, mathematics and science to technical problems or projects
9	c. identify the characteristics of, analyse and solve technical problems

Course Outline

Topic	LH	PH
Sys Admin Overview		
Operating system [preparing for installation]	3	3
Disk management and file systems.	3	3
Managing user accounts, roles and groups	3	3
Configuring networks, LAN connectivity & troubleshooting.	3	3
Network Services	3	3
Printer management: Local and network printing; Internet printing; Current trends of internet faxing	3	3
Business Continuity Planning & Disaster Recovery Planning.	3	3

Disaster protection and recovery.	3	3
Backup strategies, ICT policy formulation	3	3
Troubleshooting	3	3
Total	30	30

Mode of Delivery

The teaching style will be FACILITATOR with students being broken up into discussion groups after each major topic. 2 contact hours and 2 tutorial hours will be delivered

- Lectures: 30 LH
- Practicals 30 PH

Mode of Assessment

1. Theory test & Assignment 20%, Practical test 20%,
2. Final Exam 60%

Basic Reading List

1. Mark Burgess (2004) Principles of Network and System Administration. Published by Wiley and Sons 2nd Edition. ISBN 0-470-86807-4
2. Analytical Network and System Administration: Managing Human-Computer Systems Mark Burgess ISBN: 978-0-470-86100-4
3. "Essential System Administration", 3rd Edition, by Aileen Frisch ISBN: 0-596-00343-9,
4. "The Practice of System and Network Administration", by Thomas A. Limoncelli, Christine Hogan, Strata R. Chalup. ISBN: 0321492668
5. Linux System Administrator's guide by Lars Wirzenius
6. MCSE/MCSA Managing and Maintaining windows Server 2003 Environment Study Guide by Anil Desai <http://oreilly.com/catalog/samba/chapter/book/index.html>

Course Name	Information and Cyber Security
Code	CS3102

Level	Year 3 Semester 1
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Course Description

This course will provide a comprehensive introduction and study into a broad selection of contemporary information systems security issues, concepts and policies, including the survey of state-of-the art technology used to address security problems. Topics of study include core security mechanisms of access control and integrity, basic cryptography techniques, assurance, authentication, digital signatures and database security mechanisms. The course discusses security policy and privacy issues for information systems protection and detection, as well as an overview of recent trends in commercial products and applications and security research in basic network security, intrusion detection. Terms, technologies as they relate to NSTISSC security model are discussed.

Course Objectives

At the end of this course student should be able to;

- a) Explain the concepts relating to protection of information systems against unauthorized access to or modification of information in its various forms.
- b) Understand how to protect against the denial of service to authorized users, including measures necessary to detect, document, and counter such treats.
- c) Develop an understanding of technical terms and knowledge about techniques and methodologies for data protection and information assurance in a wide range of information systems.
- d) Develop an understanding in the differences in levels of security in the applications and usage of current commercial security products available to businesses and how they adhere to an enterprises security policy and requirements as well as the ethical and privacy issues.

Learning Outcomes/ Goals

At the end of this course student are able to:

- 1. Understand what are the common threats faced today,
- 2. Explain foundational theory behind information security,
- 3. Describe the basic principles and techniques when designing a secure system,

4. Assess the threats, and how to gauge the protections and limitations provided by today's technology.

Course Outline

Topic	LH	TH	PH
Overview of e-security - Threats, risks, consequences - Sources of threats - Attacks classification - Preventive measures, remedial measures	6	0	4
Cryptography for e-security - Historical perspective - Confusion vs. diffusion - Stream ciphers vs. block ciphers - Keys and key management - Key exchange (peer to peer, peer - keyserver - peer) - Diffie Helman key sharing scheme - Symmetric key cryptography vs asymmetric key cryptography - Trapdoor functions	4	3	6
Hash digests - Properties of cryptographic hash functions - Merkle Damgard construction, etc	2	3	2
GPG - Overview of GPG - Commands and CLI, etc	3	0	2
Elementary number theory - Prime numbers, Factoring - Modular arithmetic, etc	2	3	2
Public key encryption - Public key crypto systems - RSA algorithm - Elliptic Curve cryptography	4	0	4
Practical applications - PKI, CA. X509 certificates - SSL/TLS, HTTPS - IPV6 and IPSEC	3	3	4
	24	0	24
Total hours	60		

Mode of Delivery

Utilize a question/answer methodology (Socratic method of teaching) in which Lecture, Slides, Handouts, case studies, current news articles and assignments and used to initiate discussion in which students themselves do problem solving to arrive at conclusions about topics.

Blackboard is used to broadcast information and class updates. Use break out group sessions as well as to build communication and team working skills as well as student presentations to develop individual communication skills.

Mode of Assessment

Assessment	mark
Course works	
• Exercises	40%
• Quizzes	
• Tests	
Final exams	60%

Basic Reading List

1. William Stallings, Cryptography and network security, Pearson Education.
2. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone , Handbook of Applied Cryptography, CRC Press.

Course Name	Research Methods in Computing
Code	CSC2205

Prerequisite	None
Level	Year 2 Semester 2

Course Description

This course will enable students to identify and apply appropriate research methodology in order to plan, conduct both basic and applied research in an organization. The course will further more enable students to distinguish between scientific method and common sense knowledge, while laying the foundation for research skills at higher levels.

Course Objectives

At the end of this course, students will:

1. Demonstrate an understanding of and applying basic research concepts
2. Demonstrate an understanding of the ethical dimensions of conducting applied research
3. Identify various sources of information
4. Plan and conduct IT related research
5. Demonstrate an understanding of and applying basic descriptive statistics and qualitative analysis
6. Demonstrate an understanding of writing a basic research report of limited extend

Course Outline

Topic	LH	PH
Introduction to research._what is research, research in business, types of business research, managers and research and applying scientific thinking	3	
Scientific Investigation. Hallmarks of scientific research, Scientific research in the management area, The building blocks of science in research and Other types of research	3	
<u>Technology and Business Research</u> Information needs of business and IT, Commonly used technologies in business research, Some software used in research	3	
<u>Problem definition</u> The research process for applied and basic research, Broad problem area, Preliminary data collection, Literature survey, Problem definition	3	
<u>Hypothesis Development</u> The need for a theoretical framework, variables, theoretical framework, hypotheses development	3	
<u>Elements of research design</u> The research design, The purpose of the study, Exploratory study, Descriptive study, Hypothesis testing, Case study analysis	3	
<u>Data collection methods</u> sources of data, primary sources of data, secondary sources, data collection methods, interviewing, unstructured and structured, interviews, questionnaire, guidelines for questionnaire	3	
<u>Sampling</u> Population element, population frame and sample, Sampling, Censures and sample surveys, Probability and no probability sampling, Cluster sampling, Stratified sampling	3	

Data Analysis and Interpretation. Getting data ready for analysis, Data analysis, Data analysis and interpretation, Some software packages useful for data analysis, Summary	3	10
Concept paper, Proposal and Report writing Case Studies 1. Basic research 2. Applied Research	3	20
Total	30	30
Total CH		60

Mode of Delivery

- Lectures: 30 LH
- Practicals: 30PH. (At the end of the course a case study will be conducted where all aspects taught will be put in practice)

Mode of Assessment

- Progressive Assessment: (combined must equal 40%)
 - Assessment 1: 5%
 - Assessment 2: 20%
 - Assessment 3: 15%
- Final Exam 60%

Basic Required List

1. Uma Sekaran (2003). **Research Methods for Business--A Skill Building Approach**. John Wiley. New York. 4th Edition.
2. Kalton G (1983): Introduction to Survey Sampling. Sage Publications
3. Bryman, Alan and Duncan Cramer. 1997. Quantitative Data Analysis With
4. SPSS for Windows, Routledge, London and New York

Course Name **Computer Graphics**
Code CSC2206

Prerequisite None
Level Year 2 Semester 2

Course Description

This course exposes students to a comprehensive introduction to the fundamentals of computer graphics programming and tools. Hardware and software components of graphics systems are explored. Output and filled-data primitives. 2D and 3D geometric transformations. Two-dimensional viewing. Three-dimensional viewing: Viewing pipeline, viewing parameters, projections, viewing transformations, clipping. Visible surface detection. Introduction to illumination models and surface rendering.

Learning Objectives

The learning outcomes for this course include:

1. Describe the basic graphics pipeline and how forward and backward rendering factor in this.
2. Apply 3-dimensional coordinate system and the changes required to extend 2-dimensional transformation operations to handle transformations in 3D.
3. Explain the concept and applications of texture mapping, sampling, and anti-aliasing.
4. Implement simple procedures that perform transformation and clipping operations on simple 2D images.
5. Implement a simple real-time renderer using the OpenGL API with vertex buffers and shaders.
6. Describe the trade-offs in different representations of rotations.
7. Implement the spline interpolation method for producing in-between positions and orientations.
8. Implement basic physics-based animation algorithms for particle dynamics using simple Newtonian mechanics with Euler methods.

Topic	LH	TH	PH
Computer Graphics Overview: Images, Models, Pipeline architecture	2		2
Modern OpenGL Programming: OpenGL pipeline, Libraries, Conventions, Basic shaders.	2		4
Linear Algebra and Geomentry Reviews: Vectors, Matrices, Coordinate systems, Curves, Surfaces, Implicit vs. Parametric forms, Polygons	4		
Transformations: Points vs. Vectors, Homogeneous transformations	2		2
Viewing: Projections, OpenGL camera control	2		2
Shading: Lighting models, Surface normals, Phong shading	2		2
Texture Mapping: Mapping methods, Texture coordinates,	2		2
Buffers and Blending: Translucency, Compositing images, Anti-aliasing	2		2
Clipping: Cohen-Sutherland, Bounding boxes, Hidden-surface removal	2		2

Rasterization: Line drawing, Polygon fill, Line scanning	2		2
Maps: Environment and Shadow Maps - Reflection models, Sphere map, Cube map, Projective shadows, Depth maps, Shadow test	2		2
Animation: Camera motion, Splines, Quaternions, Dynamics, Particles, Rigid-bodies, Fluid simulation	2		4
3D Displays: Visual cues, Stereoscopic displays, Head-coupled rendering	2		2
Applications and tools: Adobe Photoshop etc	4		4
Total hours	30		30
TOTAL CH	60		

Mode of Assessment

5% -- problem sets,
5% -- online quizzes,
15% -- Tests,
15% -- course project, and
60% -- final examination.

Course Material

Require Text

- “Interactive Computer Graphics” 7th Edition by Edward Angel (Pearson, 2012).

Recommended Text

- “OpenGL Programming Guide” 8th Edition by Dave Shreiner et al. (Addison-Wesley, 2013).

Course Name **Web Applications Development**
Code **CSC2208**

Prerequisite None
Level Year 2 Semester 2

Course Description

Comprehensive course in Python, an open-source dynamic object-oriented scripting language. Covers variables, arrays, lists, tuples, dictionaries, functions, methods, classes, objects, and writing server-side Python scripts for the Web. Also covered are exception handling, regular expressions, and modules. An introduction to SQL and the MySQL database, and advanced topics such as Model-View-Controller architecture and Web application development with the Django framework. Intended for students with previous programming experience.

Course Objectives

Upon successful completion of this course, a student will be able to:

1. Develop server-side Python scripts for publishing on the Web.
2. Employ control structures, functions, and arrays to create Python programs.
3. Apply object-oriented programming concepts to develop dynamic interactive Python applications.
4. Employ Python sequences and mappings to store and manipulate data.
5. Use SQL commands and the MySQL database together with Python.
6. Create an advanced project using MySQL, Python and a Model-View-Controller framework.

Learning Outcomes/ Goals

Upon successful completion of this course, a student will meet the following outcomes:

1. Develop server-side Python scripts for publishing on the Web.
2. Employ control structures, functions, and arrays to create Python programs.
3. Apply object-oriented programming concepts to develop dynamic interactive Python applications.
4. Employ Python sequences and mappings to store and manipulate data.
5. Use SQL commands and the MySQL database together with Python.
6. Create an advanced project using MySQL, Python and a Model-View-Controller framework.

Course Outline

Topic	LH	TH	PH
Introduction • History of Python • Dynamic Web Applications • Python Installation • Web Servers and Server-Side Programming • Python and Databases • f. UNIX/Linux Environment	3	0	3

Data and Operations • Data Types • Variables and Constants • Operators d. Expressions • Operator Precedence • f. Scope	2	0	2
Conditional Statements and Control Structures • If Else • Elif Clause • While • For • Parallel Iteration • f. Numbered Iteration	2	0	1
Functions • Function Definition • Blocks c. Calling a Function • Function Parameters • Function Return Values • Lambda Expressions • Modules • h. Recursion etc.	3	0	3
Collections • Sequences ○ Lists ○ Strings ○ Tuples • Mappings (Dictionaries) • Array Module	2	0	3
Object-Oriented Python • Defining a Class • Constructors • Encapsulation • Inheritance • Interfaces • Iterators • Generators • Polymorphism, etc.	3	0	2
Web Interaction and File Access • Python HTML Code Generation • CGI Code Generation	2	0	2

• Templating Systems • Form Input • Validation and Regular Expressions • Cookie Module			
SQL and the MySQL Database • Introduction to SQL Syntax • Designing and Creating a Table in MySQL • MySQL Data Types • MySQLdb Module	3	0	5
Dynamic Python Applications with Django and MySQL • Model-View-Controller Architecture • Connecting to MySQL • Object-Relational Mapping • Authentication • Testing and Debugging	4	0	3
Introduction to ASP.NET Overview of the .NET Framework, A Bit of C#, Introduction to ASP.NET, ASP.NET Controls, ASP.NET AJAX, Web Services			
Database Access through the Web Relational Databases, An Introduction to the Structured Query Language, Architectures for Database Access, The MySQL Database System, Database Access with PHP and MySQL, Database Access with JDBC and MySQL, Database Access with ASP.NET and MySQL	3		3
Ruby on Rails Origins and Uses of Ruby, Scalar Types and Their Operations, Simple Input and Output, Control Statements, Fundamentals of Arrays, Hashes, Methods, Classes, Blocks and Iterators, Pattern Matching, Overview of Rails, Document Requests, Rails Applications with Databases	3		3
	30	0	30
Total hours	60		

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.

- Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

Assessment	mark
Course works	
• Exercises	40%
• Quizzes	
• Tests	
Final exams	60%

Basic Reading List

1. Shaw. Learn Python the Hard Way, online ed. <http://learnpythonthehardway.org/>, 2014.
2. Ceder. The Quick Python Book, 2nd ed. Manning Publications, 2010.
3. Zelle, John. Python Programming: An Introduction to Computer Science, 2 ed. Franklin, Beedle & Associates, 2010.
4. Lutz, Mark. Programming Python, 5th ed. O'Reilly Media, 2013.

Course Name	Distributed and Cloud Computing
Code	CS3106

Level	Year 3 Semester 1
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Course Description

This course covers Internet technology and network-based computing systems, including clouds, clusters, datacenters, grids/P2P and Internet of Things (IoT). Specific topics include server clusters, supercomputers, P2P networks, virtual machines, data/computational grids, Internet clouds, IoT, social networks, big data security, etc. Case studies include Amazon EC2 and S3, Google AppEngine, Azure, SalsForce, Facebook, Twitter, MyPlace, Rackspace, Dropbox, iCloud, XEN, KVM, Chord, BitTorrent, MapReduce, Hadoop, Eucalyptus, OpenStack, VMWare Tools, etc.

Course Objectives

The objective of this course is:

- a. Design and implement distributed and cloud systems.
- b. Understand various basic concepts related to cloud computing technologies
- c. Understand the architecture and concept of different cloud models: IaaS, PaaS, SaaS
- d. Understand big data analysis tools and techniques
- e. Understand the underlying principle of cloud virtualization, cloud storage, data management and data visualization.

Course Learning Outcomes

Upon successful completion of this course you should be able to:

- a. Develop and deploy distributed and cloud applications using popular platforms,
- b. Design and develop highly scalable cloud-based applications by creating and configuring virtual machines on the cloud and building private cloud.
- c. Explain and identify the techniques of big data analysis in cloud.
- d. Compare, contrast, and evaluate the key trade-offs between multiple approaches to cloud system design, and Identify appropriate design choices when solving real-world cloud computing problems.
- e. Make recommendations on distributed and cloud computing solutions for an enterprise.

Overview of Learning Activities

The learning activities included in this course are:

- a) Key concepts and basic principles will be explained in lectures with current industry examples of applications and solutions.
- b) Tutorials and labs will help students learn the tools and techniques, such as how to use cloud computing tools offered by industry leaders such as Amazon, Google, Microsoft and practice report writing by analysing case studies.
- c) Project-based learning, where students will work in a group to analyse and solve industry-related problems with cloud computing solutions
- d) Private study, working through the course as presented in classes and learning materials, group discussions (including online forums), and gaining practice at solving conceptual and technical problems.

Methods of Delivery

This course will be offered in forms of:

- Case studies
- Lectures
- Projects and student centered learning activities
- Activity Labs

Methods of Assessment

The course will be assessed using a variety of methods including in-class exercises, assignments, quizzes, Course project and a written exam. Summary of method coverage is indicated below.

No.	Item	Coverage (%)	Frequency
1	Class activity and assignments with presentations	5	3
2	Projects	15	1
3	Quizzes	10	2
4	Mid terms	30	2
5	Final Exam	40	1

Detailed Content

Topic	Sub topic / Labs required	Contact Hours
Course introduction		3
Introduction to cloud and distributed computing	Introduction to Amazon Web Services - Launching an EC2 Machine Instance	3
Virtual machine monitors	Creating an Amazon Machine Image (AMI)	6
Auto scaling computing clusters	Elastic load balancing and autoscaling	3
Developing and deploying clustered Web applications	Deploying a web application	4
Distributed file systems		3
Non-relational database systems	Non-relational database application, SimpleDB	4
Midterm		1
Data intensive processing in large clusters. Google's MapReduce. Apache's Hadoop	MapReduce	3
Warehouse scale computing architecture		3
Map reduce and DFS applications		3
Map reduce and databases		3
Map reduce graph processing and PageRank		2

Distributed retrieval and data mining		2
Final Exam (2 parts: Limited theory and Practical project).	Final project	6
Total		48

References and materials

Materials

1. Hwang, K., Dongarra, J. and Fox, G.C., 2013. *Distributed and cloud computing: from parallel processing to the internet of things*. Morgan Kaufmann
2. Bagchi, S. ed., 2015. *Emerging Research in Cloud Distributed Computing Systems*. IGI Global.
3. Buyya, R., Vecchiola, C. and Selvi, S.T., 2013. *Mastering cloud computing: foundations and applications programming*. Newnes.

References

1. Di Costanzo, A., De Assuncao, M.D. and Buyya, R., 2009. Harnessing cloud technologies for a virtualized distributed computing infrastructure. *IEEE Internet Computing*, 13(5), pp.24-33.
2. Zhang, S., Zhang, S., Chen, X. and Huo, X., 2010, January. Cloud computing research and development trend. In *2010 Second international conference on future networks* (pp. 93-97). IEEE.
3. Jonas, E., Pu, Q., Venkataraman, S., Stoica, I. and Recht, B., 2017, September. Occupy the cloud: Distributed computing for the 99%. In *Proceedings of the 2017 Symposium on Cloud Computing* (pp. 445-451). ACM.

COURSE NAME: Citizenry, Professionalism, Globalization and Entrepreneurship

CODE: DVS 2201

COURSE LEVEL: Degree Year 1 Semester 2

CREDIT UNITS: 3

CONTACT HOURS: 45

Pre requisite: None

Description

Effective citizenship is a very important deciding factor in nation building in both developed and developing countries. Also, national development plans (economic, social, political, scientific and technological aspects) cannot be well implemented without the contribution of effective and incorruptible citizens. This course looks at the present situation in Uganda

society and posits that the nation cannot develop or make necessary progress without effect citizens who are incorruptible and committed. One of the challenges facing the public as well as private sector is professionalism. Increasingly, institutions have hired professionals, but this has not improved the quality of service; the quality of service has slowly gone down. The current paradigm of new public management with emphasis on market mechanisms in the delivery of public service comes in as a redress of public failure to deliver service. Within the New Public Management framework the private –public partnership ideals have been introduced in the public service sector. Despite these approaches, service delivery continues to be challenging; there is increased outcry over corruption with increased number of cases reported from time to time. This course unit examines professions and the extent to which professionalism is practiced in Uganda.

This course explores the socioeconomic implications of globalization and some fundamental changes that have taken place in the global socioeconomic system. This course examines the economic, social, political, environmental, and cultural dimensions of globalization from African perspectives. Major global issues such as democratic governance, inequalities of development, poverty eradication, conflict and war, disease control, climate change, financial crises, natural resource exploitation, and diasporization, cultural change, and appropriation will be illustrated and analyzed from the perspective of Africa's experience of globalization. Furthermore the course aims at examining the nature of new business opportunities in Africa, to gain insights on how a new generation of African entrepreneurs is exploiting these opportunities and meeting the increasingly complex entrepreneurial challenges as Africa merges into the mainstream of global commerce.

Course Objectives

The objectives of this course are:

- a) To teach students key concepts in Citizenry, Professionalism and Development.
- b) Make students understand opportunities and challenges of globalization.
- c) To explain the role of entrepreneurship in the development of Uganda and Africa
- d) To teach ethics and integrity for professional development.

Learning outcomes

- a) Identify and describe the major global issues confronting Africa today

- b) Identify and analyze the economic, political and socio-cultural implications of foreign investment in Africa
- c) Understand the role of entrepreneurship for Africa's development
- d) Explain the relevance of professional ethics for society's development

The outline is divided into three major sections

1. Citizenry, Professionalism and Development (15 Hours)

- Definition and Key concepts: Ethics, Morality, Integrity
 - Theories of ethics- Consequentialism and Non-consequentialism; Deontological Theory, Virtue Theory
 - Professional ethics
 - Responsible citizenship
- civil ethics and professional ethics, responsible citizenship (patriotism) –Institutional trust (trust in Public institutions), self-awareness, confidence and community participation, corruption and the development agenda.

2. Globalization and Development (15 Hours)

- Critical and historical approach to Globalization
- Contemporary Globalizing processes (Cash flows, State M-Market Relations, Transnational Corporations, Global Commodity Chains, Inequality and Poverty)
- Labor Migration
- Globalization and Imperialism (Post-Cold War Imperial and civil wars, Global and regional Challengers to US Hegemony: Russia and China)
- Globalization Democracy and Culture (Human Rights, Democratization, cosmopolitanism, standardization and homogenization)
- Foreign Aid, Grants/Donations: Implications on Sustainability
- Alternative to Neo-liberal Globalization (Global Labor Movement, Transnational Social Movements, NGOs and Environment)
- New Economic Arrangements: The fall of USSR and the triumph of Capitalism
- Emerging Economies: the BRICS and the rise of China as a super power , The NICs (Asian Tigers)

- Modern Slavery and its Effects- Labor, Human Trafficking, Terrorism and Religious Fundamentalism

3. Entrepreneurship and Development in Africa (15 hours)

- Entrepreneurship and management, entrepreneurship skills development, career opportunities in Entrepreneurship, finance and development, basics in financial management, Uganda's policies on entrepreneurship development
- Roots of entrepreneurship: economic, cultural, social considerations
- Settings for entrepreneurship: private and public sectors; large, small, and micro enterprises
- Private enterprise and industrial development-Small Scale, medium and large scale, industries
- Entrepreneurship in comparative perspective: The nature and diversity of entrepreneurship and entrepreneurs in Africa: Emerging business opportunities
- Competencies and skills are necessary to succeed in an African business context
- Challenges and innovation in business spheres
- The role of leadership in entrepreneurship

Mode of delivery:

- Lectures
- Case studies and class exercises.
- Group assignments and presentations.
- Group reading.

Mode of Assessment:

No	Mode of Assessment	Mark (%)
1	Mid-semester exam or take home assignment	30
2	Class attendance and Students presentations	10
3	Final Examinations	60
	TOTAL	100

Instructional Materials

13. LED Projectors

- 14. Laptops
- 15. Teaching boards and writers
- 16. Rooms with enough chairs and tables.

Reading Material

George Ritzer and Paul dean (2018) Globalization: The Essentials 2nd Edition. Wiley-Blackwell

Mamdani, Mahmood. (1996). Citizens and Subject: Contemporary *Africa and the Legacy of Late Colonialism*. Princeton, NJ: Princeton University Press

Steve Mariotti (2014) The Young Entrepreneur's Guide to Starting and Running a Business: Turn Your Ideas into Money

Course Name: **Internship**
Code: CS3214

Level: Year 3 Semester 2 Recess

Course Description

Practical Training in public or private firms that utilize technology for a period of eight weeks or more. Here, a student is expected to acquire hands on training in computing or related field. This experience gives students an opportunity to integrate theory and practice by working in a supervised setting. The supervision is done by both the field and the academic supervisors.

Course Objectives

Upon completion of this training, the student will submit an industrial training report, detailing all that he/she learnt during the training/internship. At completion of this experience, the student should be able to:

1. Integrate and apply theory and understandings from CS courses in a work environment,
2. Observe and take part in the management of technology changes within an organization,
3. Understand the industry environment of the organization providing the internship,

4. Understand the impact and role of computers and information technology on an organization,
5. Demonstrate community involvement and reflect on the issues encountered in that involvement,
6. Demonstrate the ability to work with supervision and with other employees, and
7. Perform various tasks to the satisfaction of the supervisor.

Learning Outcome

1. Upon completion of this training, a student will submit an industrial training report, detailing all that he/she learnt during the training/internship.

Internship Duties

Students should keep an informal journal of activities and impressions throughout the internship experience. At the mid-point of the experience, a summary report of those experiences is due which gives an overview of the typical daily or weekly activities including: tasks performed, interactions made with employees, meetings attended, and the overall atmosphere of the work environment. Identify the positive and negative things about the internship along with observations and reservations made about the job or career. Note any new skills, knowledge, or insights acquired or new perspectives on older skills, knowledge, or insights.

Evaluation

A typical 4 credit hour internship consists of at least 8 weeks of internship experience after during the third-year recess semester.

Grading is based on the marks awarded by the field supervisor and the marks awarded by the academic supervisor and the report marks.

The student will submit **two** written reports and maintain regular contact with the faculty advisor.

4.5 Year 3 Semester 1

Course Name	Software Project Management
Code	CS3101

Level

Year 3 Semester 1

Course Description

This course is to equip the students with the basics in Project Planning and Management with focus on IT Projects. It looks at overview of general management and the ten project management knowledge areas. The course also touches on monitoring and evaluation of projects. The module will define the nature of projects and management. Various aspects will be considered including financial control and management of risks, human resource elements (building and leading a multi-disciplinary team) and the control of time. This module will examine the integration of time, cost and quality aspects of projects through the application of relevant tools and techniques. Project management is an essential skill in today's business world.

2. Course Objectives

At the end of this course student should be able to:

- i. Understand the meaning of management and its functions
- ii. Understand the importance of Project planning and management in projects
- iii. Identify and explain project life cycle and explain the project's process groups
- iv. Develop WBS for a simple IT Project and Schedule it using Gantt and Network diagrams
- v. Explain ethical, integrity, and professionalism in project management

3. Learning Outcomes

At the end of this course student are able to:

- i. Explain general management and management functions
- ii. Explain the importance of Project planning and management in projects
- iii. Identify and explain the processes involved in the 10 project management knowledge areas
- iv. Identify an IT Project, develop a WBS and Schedule it using Gantt and Network diagrams

- v. Explain the importance of ethical, integrity, and professionalism in project management

4. Methods of delivery

1. Lectures
2. Group discussions
3. Assignments
4. Practicals

Course Duration (Hours)	Credit Units	Year when offered
60	3	Year 3, Semester 1

5. Methods of Assessment

No	Assessment	Marks %
1	Coursework	40
2	End of Semester Examination	60
	Final Mark	100

6. Course Outline

Topic	Sub Topics	Contact Hours
Introduction Concepts	• The management functions, Roles and Skills • Projects, Programs and Operations • Why Projects come up • Critical Success and Failure Factors in Projects • Indicators of Successful and Failed Projects	4
Project and Project Management	• Characteristics of a project	4

	• Project Life Cycle and its phases • Project Planning and Management • Introduction to Project Management Knowledge Areas • The Project major Constraints	
Project Management Process Groups	• Initiating process, • Planning process • Executing process • Controlling process • Closing process	4
Project Planning and Management Knowledge Areas (Input, Tools, Techniques, and Output of each.)	• Project Integration Management • Project Scope Management • Project Communication Management • Project Human Resource Management • Project Procurement Management • Project Quality Management • Project Risk Management • Project Cost Management • Project Time Management • Project Stakeholders Management	26
Project Scheduling	• Precedence diagramming method (PDM ,AON),	2

	• Arrow Diagramming method (ADM, AOA), • Early Start, Early Finish, Late Start, Late Finish of activities • Activity Floats/Slacks • Critical activities and Critical path of a project	
Application of Microsoft Project	• Gantt Charts • Network diagrams • Calendars • Resources	2
Project monitoring & Evaluation	• Monitoring Vs Evaluation • What to monitor in projects • Importance of Monitoring and Evaluation • Types of Evaluation • The Project Evaluation Process	4
Professional and Ethical Issues in PPM	• Ensuring Personal Integrity • Ensuring Product Integrity	2
Total Contact Hours		48

7. Reference List

2. Kim Heldman, Project Management Professional Study Guide, Third Edition, Wiley Publishing House, 2006, USA.
3. Blackman, M. Project cycle management. Tear fund 2003. ROOTS resources

4. Dobie, C. (2007). A handbook of Project Management: A Complete Guide for beginners to professionals, New SouthWales: Allen and Unwin.
5. Project Management Institute. (2008). A Guide to the Project Management Body of Knowledge (PMBOK) 4th Edition, Project Management Institute, Upper Darby, PA
6. Project Management Institute. (2013). A Guide to the Project Management Body of Knowledge (PMBOK) 5th Edition, Project Management Institute, Upper Darby, PA.
7. Cuthworth, J.W. and Franks, T. (1993). Managing Projects in Developing Countries, Prentice hall, London

Course Name **Artificial Intelligence**

Code CS3110

Level Year 3 Semester 1

Course Description

This course provides an overview and introduction to the field of Artificial Intelligence. It introduces students to the basic knowledge representation, problem solving, and learning methods of artificial intelligence.

Notions of rational behaviour and intelligent agents will be discussed. Major subareas will be covered. The emphasis will be on understanding the fundamental concepts, as well as being able to practically apply the corresponding approaches in solving practical problems and developing useful software applications. Program illustrations in traditional languages such as C and Java will be used.

Course Objectives

At the end of this course student should be able to:

- a) Explain strong foundation of fundamental concepts in Artificial Intelligence
- b) Understand the notions of rational behaviour and intelligent agents.
- c) Develop a general appreciation of the goals, subareas, achievements and difficulties of AI.

- d) General an understanding of major concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.
- e) Developing programming skills for AI applications.

Course Outline

Topic	LH	PH
Introduction to Artificial Intelligence	2	
Intelligent agents and expert systems	4	3
Problem solving through uninformed search	2	
Problem solving through informed search. Heuristics	2	
AI and playing games	2	3
Introduction to knowledge representation	2	
Knowledge representation. Logical reasoning.	2	2
Propositional logic	2	4
Prolog programming	4	8
Predicate	2	4
Introduction to Robotics	4	8
Present and future of AI	2	4
Total	30	30

Mode of Delivery

Lectures and practical

Mode of Assessment

The assessment for this course is:

•Assignments	10 %
•Labs	10 %
•Tests	20 %
•Final Examination	60 %

Basic Reading List

1. Required textbook: S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach, Third Edition*, Prentice Hall, 2009 (or 2010). Available at the McGill bookstore.
2. Nils J. Nilsson, "Artificial Intelligence: A new Synthesis", Harcourt Asia Pvt. Ltd., 2000.
3. Elaine Rich and Kevin Knight, "Artificial Intelligence", 2nd Edition, Tata McGraw-Hill, 2003.

4. George F. Luger, "Artificial Intelligence-Structures and Strategies for Complex Problem Solving", Pearson Education / PHI, 2002.

Course Name	Embedded Systems
Code	CS3104

Level	Year 3 Semester 1
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Course Description

Embedded systems have become the next inevitable wave of technology, finding application in diverse fields of engineering. Microprocessors, together with sensors and actuators, have become embeddable in almost everything. The purpose of the course is to provide the students with the basic information about embedded systems which can be defined as a control system or computer system designed to perform a specific task. The course consists of two parts being closely interconnected. The aim of the course, in its first half, is to introduce students to the theory and practice of control system engineering. The second part is devoted to the basis of microcontrollers architecture and programming of embedded systems.

Introduction to designing microcontroller-based embedded computer systems using assembly and C programs. Examination of Real-time Operating Systems and their impact on performance. Computer engineering applications will be emphasized.

Course Objectives

The goal of this course is to solidify and build upon a student's knowledge of computer organization by presenting hands-on experience with microcontrollers. Students will also examine a few sensors that are used in commercial and medical products and learn how to interface them in a microcontroller system. Students will:

1. Develop modern microcontroller in assembly language and operate its peripheral devices.
2. Describe how the assembly code generated by a compiler relates to the original C code.
3. Implement embedded systems using different peripheral devices, input, and communications devices.
4. Understand thread-based program design with a real-time operating system.

5. Develop programs controlling embedded systems using quick and efficient methods.

Learning Outcome

Students should be able to demonstrate the following competencies and knowledge:

1. Design an embedded system (including hardware and software) that meet a written set of requirements.
2. Implement an embedded system (including hardware and software) that solves a real world problem using engineering processes.
3. Use hardware and software development techniques, skills, and computer tools to solve a real-world problem.

Prerequisite

Course: CS2207 (Digital Electronics and Logic Design) or departmental approval

- Knowledge of C, C++, Java, or other high-level programming
- Knowledge of complex digital circuits, including computer architectures
- Knowledge of analog circuits
- Knowledge of modern computer-based design tools

Course Content

Topic	LH	TH	PH
Introduction to Embedded Systems and Microcontroller-based Circuit Design	2		4
Instruction Set Architecture	2		4
C Programming Review	2		4
C Start-Up Module and Digital I/O	2		4
Analog to Digital Conversion	3		4
Disciplined Software Development	2		
Serial Communications	3		
Interrupt concepts and behavior with C programming	2		4
Interrupt-Driven Serial Communications and Data Sharing	2		4
Scheduling	2		4
Software Testing	2		4
Process Coordination	2		
Total hours	26		34
TOTAL CH	60		

Mode of Delivery

- 12 Lectures throughout the semester.

- In class and out of class labs and practical

Mode of Assessment

5% -- problem sets,
15% -- Tests and quizzes,
20% -- course project, and
60% -- final examination.

Course Material

Microcontroller

- Aduirno board and accessories
- Renesas RX63N Microcontroller

Require Text

- Introduction to Embedded Systems using the Renesas RX63N Microcontroller.
- Beginning Arduino Programming by Brian Evans, APress

Recommended Text

1. E. A. Lee, S. A. Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, Second Edition, ISBN 978-1-312-42740-2, 2015.

Course Name	Computational Complexity
Code	CSC3110

Prerequisite	None
Level	Year 2 Semester 2

Course Description

Formal aspects of describing and recognizing languages by grammars and automata. Parsing regular and context-free languages. Ambiguity, nondeterminism. Elements of computability; Turing machines, random access machines, undecidable problems, NP-completeness.

Learning outcomes

The students will be able to:

1. Describe the main features of the proofs of important theorems;
2. express problems from relevant areas of applications in a mathematical form suitable for further analysis;
3. Use the theory, methods and techniques of the course to solve mathematical problems;
4. Solve problems of greater complexity, i.e. problems requiring a combination of ideas and methods for their solution, and be able to give a more detailed account of the proofs of important theorems and by examples and counter-examples be able to motivate the scope of various results.
5. Present mathematical arguments and reasoning will be greater.

Course Outline

Topic	LH	TH	PH
The notion of an algorithm.	2		
Machine models for computations: variants of Turing machines and their equivalence.	2	3	2
Complexity theory: complexity hierarchies, effective reducibility, the $P=NP$ problem, NP-complete problems, complexity of various decidability problems in logic and automata theory, Savitch's theorem.	8	3	2
The polynomial hierarchy.	2		2
Boolean functions and circuits.	2	3	
First and second order logic complexity and their relation to complexity classes.	2		2
The relativized $P=NP$ -problem.	4	3	
Probabilistic algorithms and their complexity theory.	2		2
Lower bounds for proof lengths in systems of proposition logic.	2	3	
Complexity for other computational models, such as exact real arithmetic, higher type functionals, and quantum computations.	4	3	2
Total hours	30	18	12
TOTAL	60		

Mode of Assessment

Assessment will consist of:

Weekly Assignments	20%
Tests	20%
Final exam	60%

Reading List

1. Michael Sipser, Introduction to the Theory of Computation, 3rd Edition
2. John Hopcroft, Rajeev Motwani, Jeffrey Ullman, Introduction to Automata Theory, Languages, and Computation, 3rd edition. Wesley, 2007

Course Name **Mobile Applications Development**
Code **CS3109**

Level Year 3 Semester 1

Course Description

Students learn how to develop applications for mobile devices, including smartphones and tablets. Students are introduced to the survey of current mobile platforms, mobile application development environments, mobile device input methods, as well as developing applications for two popular mobile platforms. Students will design and build a variety of Apps throughout the course to reinforce learning and to develop real competency.

Course Objectives

At the end of this course student should be able to:

- a) Develop apps for iOS platform devices (iPhone/iPad/iPod Touch)
- b) Understanding of computer architecture and object-oriented programming
- c) Develop a working knowledge of Apple's XCode app development tool
- d) Understand mobile design principles

Learning Outcomes/ Goals

Upon successful completion of the course, the student will demonstrate the ability to:

- a) Explain mobile devices, including their capabilities and limitations.
- b) Use current mobile platforms and their architectures.
- c) Develop mobile applications on a popular mobile platform.
- d) Evaluate development with another mobile platform.

Course Outline

Topic	LH	TH	PH
Mobile Phones and Network Technologies	1	0	1

Android Software Development			
– Overview, The Tools, The Architecture of Android Applications, The Execution Model for Android Applications, View Groups, Simple Views, Running an Application on an Android Device, Using the Class to Call Other Activities, Widgets, Dealing with Lists, Data Persistence, Using the Debugger	2	0	2
Android Application Frameworks	1	0	1
Building a Simple User Interface	2	0	2
Activities and Intents	1	0	1
Case Study: Calculator App. – Design Challenges	2	0	2
Services	1	0	1
Broadcast Receivers	1	0	1
Data Persistence	1	0	1
Processes and Threads	1	0	1
Asynchronous Tasks	2	0	2
Internet Resources	1	0	1
App Publishing and Business Models	1	0	1
Introduction to iOS platform	1	0	1
Application Development in iOS	2	0	2
Objective-C	2	0	2
Building a Simple User Interface in iOS	2	0	2
Total hours	24	0	24
	48		

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.
- Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

Tests and Quizzes	10%
Assignments	10%
Course Project	20%
Final Exam	60%
Total	100%

Basic Reading List

1. Bill Phillips, Chris Stewart, Brian Hardy, and Kristin Marsicano, Android Programming: The Big Nerd Ranch Guide, Big Nerd Ranch LLC, 2nd edition, 2015.
2. Christian Keur and Aaron Hillegass, iOS Programming: The Big Nerd Ranch Guide, 5th edition, 2015.
3. Valentino Lee, Heather Schneider, and Robbie Schell, Mobile Applications: Architecture, Design and Development, Prentice Hall, 2004.

Course Name **Computer Vision & Image Processing**
Code **CS3111**

Level Year 3 Semester 1

Course

This course will cover methods in image processing and computer vision, with an emphasis on techniques currently used in academia and industry. Topics will include image filtering, edge detection, corner detection, segmentation, object/image/face classification, object detection, morphological operators, object tracking, camera calibration, image registration, and activity classification. Students are expected to have some familiarity with calculus, linear algebra, and basic probability and statistics (conditional probability, mean, variance, etc.). Basic skills in computer programming is required.

Course Objectives

By the end of this course, students will be able to;

- a) Understand how to remove noise and restore corrupted images without distorting them

- b) Apply different techniques to represent and approximate images so as to extract meaningful components from them with respect to a particular application
- c) Segment an image into regions of interest that possess homogeneous characteristics while targeting a semantic interpretation of the image content.
- d) Apply different detection and machine learning based algorithms to image datasets so as to deduct a number of conclusions basing on the patterns learnt in the data itself.
- e) Develop applications using computer vision techniques.

Learning Outcomes

At the end of the course, a student should be able to;

- a) Identify the methods, concepts, and terminologies used in the field of computer vision and image processing
- b) Process the image datasets to analyze the different segments of interest.
- c) Design and implement various algorithms for digital image processing and computer vision
- d) Provide practical solutions to problems involving image processing tasks such as detection and tracking of object(s) of interest in a in video stream scene

Methods of Delivery

- 1) Lectures
- 2) Tutorials
- 3) Hands-on Projects

Methods of Assessment

No.	Assessment Tool	Mark %
1	Coursework	40
2	End of Semester Examination	60
	Final Mark	100

Detailed Content

Topic	Sub Topics	Contact Hours
Introduction	Overview and Progress of Computer vision, computer imaging systems	06
Image formation and Transformations	Coordinate Transformations	06

	Perspective, Pin hole Camera Model, Cognitive aspects of color, illumination: Sampling and Quantization	
Image analysis and pre-processing	Enhancement of the image by Noise Removal, Blurring, Contrast, Histogram Modification, Stretching, Filtering; Morphological filtering, Median and the Gaussian Filters	06
Edge detection	Canny / Gaussian/ Gabor/Texture Edges/ Curvature / Corner Detection	06
Segmentation	Background Subtraction, Texture Segmentation, Thresholding, Edge-Based Segmentation, Region-Based Segmentation	06
Feature extraction	Image feature, shape, histogram, color, spectral, texture. Algorithms such as Histogram of oriented gradients (HOG), Local binary patterns (LBP), Haar wavelets, Color histograms.	06
Feature Selection	Techniques in feature selection: Principal component analysis (PCA), Kernel PCA. Areas of application of the techniques discussed	06
Supervised Machine learning	Support Vector Machines, Neural Networks, Naive Bayes, Decision Trees	06
Unsupervised machine learning	Clustering: K-means Clustering	06
Applications: Object detection	A practical project on Face Detection and Recognition	06
Total Contact Hours		60

References

1. Computer Vision and Image Processing: Fundamentals and Applications, Manas Kamal Bhuyan, 2019
2. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2010
3. Computer Vision: A Modern Approach: D. Forsythe and J. Ponce, Prentice-Hall, 2003
4. Milan Sonka, Vaclav Hlavac and Roger Boyle, Image Processing, Analysis, and Machine Vision. Brooks/Cole / Thomson 1999 Basics of some image processing aspects. Texture
5. Digital Image Processing, Prentice Hall, Rafael Gonzalez, and Richard Woods, 2008
6. Digital Image Processing - R.C.Gonzalez & P.Wintz
7. Fundamentals of Digital Image Processing - A.K. Jain

Course Name: Wireless and Mobile Communications

Course Code: CSC3105

COURSE LEVEL: Year Three, Semester One

Course Description

The intention of the course is to help students to examine fundamental concepts of mobile cellular communications and specifics of current and proposed cellular systems, understand the core issues encountered in the design of wireless (vs. wired) networks. It also exposes students to fairly recent paradigms in wireless communications. Topics include frequency reuse; call processing; propagation loss; multipath fading and methods of reducing fades; error correction requirements and techniques; modulation methods; FDMA, TDMA, and CDMA techniques; microcell issues; mobile satellite systems; GSM, cdmaOne, GPRS, EDGE, cdma2000, W-CDMA, LTE and candidate 5G waveforms.

Course Objectives

By the end of this course, students will be able to;

- a. Understand fundamental concepts of mobile cellular communications and specifics of current and proposed cellular systems.

- b. Synthesis and analyze wireless and mobile cellular communication systems over a stochastic fading channel
- c. Understand digital cellular systems (cdma2000, W-CDMA, LTE and candidate 5G waveforms).

Learning Outcomes

At the end of the course, a student should be able to;

- a. Explain, analyze and design wireless and mobile cellular systems.
- b. Have the ability to work in advanced research wireless and mobile cellular programs.

Contact Hours	Credit Units	Year When Offered
60	3	3

Methods of Delivery

- 4) Lectures
- 5) Tutorials
- 6) Case Studies

Methods of Assessment

No.	Assessment Tool	Mark %
1	Coursework	40
2	End of Semester Examination	60
	Final Mark	100

Detailed Content

Topic	Sub Topics	Contact Hours
Introduction/Concepts	Introduction to wireless communications: history and evolution, current wireless communication systems, requirements of wireless services, and technical challenges of wireless communications	06
Propagation and System Planning	Radio wave propagation in the mobile environment: Free-space propagation, propagation mechanisms, large scale and	10

	small scale fading, path loss models, statistical channel models: narrowband and wideband models, System Planning: mobile radio link design, and introduction to radio network Planning.	
Wireless Access	Overview of wireless access networks: base and subscriber stations, multiple access technologies, noise and interference in wireless communication systems, diversity reception, MIMO communication: MIMO narrowband channel model, transmit diversity and spatial multiplexing	12
Cellular Systems	Evolution of cellular systems, principles and operation of cellular systems, narrowband systems: FDMA and TDMA systems, frequency planning, and capacity considerations, CDMA wideband systems: re-source allocation, soft handover, power control, interference and capacity, OFDMA wideband systems, and Standardized cellular communications systems	12
Wireless Network Standards	Wireless LANs, wireless MANs, short range wireless networks, standards, capabilities and applications, broadband wireless networks, and integration of different types of wireless networks.	10

Wireless Sensor Network	Introduction to sensor networks and applications, issues in sensor networks in comparison to conventional wireless networks, special design considerations in energy conservation, routing etc.	10
Total Contact Hours		60

References

1. Wireless Communications", A. Goldsmith, Cambridge University Press, 2005.
2. Wireless Communications", A. Molisch, Wiley-IEEE, 2nd Edition, 2010.
3. Wireless Communications: Principles and Practice", T. S. Rappaport, Prentice Hall, 2nd Edition, 2002.
4. Fundamentals of Wireless Communications", D. Tse and P. Viswanath, Cambridge University Press, 2005.
5. Articles in *IEEE Communication Magazine* and *IEEE Wireless Communication Magazine*

Course Name: Multimedia Technology

Course Code: CSC3104

COURSE LEVEL: Year Three, Semester One

Course Description

Multimedia Technologies is an indispensable part of modern computing environments. The main aim of this course is to help students develop an understanding of the fundamental principles of multimedia systems and how they are being developed and applied. The course will explain the technologies underlying digital images, videos and audio contents, including various compression techniques and standards, and the issues to deliver multimedia content over the Internet.

Course Objectives

Upon successful completion of this course, you should be able to

1. Perform a comparative analysis of the major methods and algorithms for multimedia data compression.
2. Explain the technical details of popular multimedia compression standards.
3. Write code and develop a multimedia application using JAI and JMF.

4. Explain the principles and technical details of several wired and wireless networking protocols.
5. Explain technical aspects of popular multimedia web applications, including VoD and VoIP.

Methods of Delivery

This course will be offered in forms of:

- Case studies
- Lectures
- Projects and student centered learning activities
- Activity Labs

Methods of Assessment

Assessment for this course is based entirely on a portfolio of evidence, explicitly matched to the learning outcomes of the course and including personal reflections on the learning process. Each student's work will be unique, so we encourage sharing and mutual support in this process, and allow you to such support and sharing with others as evidence that you have met the learning outcomes.

No.	Item	Coverage (%)	Frequency
1	Class activity and assignments with presentations	5	3
2	Projects	15	1
3	Quizes	5	2
4	Mid terms	15	2
5	Final Exam	60	1

Detailed Content

Topic	Sub topic	Contact Hours
Software Installation and Preparation	- What software tools are needed? - How to get started?	5
Introduction to Multimedia Studies	- What is multimedia? - Multimedia and Hypermedia	7
Data Representation	- Graphics and image data types - Popular file formats	7
Basic Compression Techniques	- Lossless compression algorithms - Loss compression algorithms	7

Video and Audio Data Compression Techniques	• Video compression based on motion compensation. • Search for Motion Vectors: <i>hierarchical, sequential, and 2d logarithmic</i>; • H.261 295- <i>inter-frame coding, intra-frame coding, quantization in H.261 technique, H.261 Encoder and Decoder, A Glance at the H.261 Video Bit Stream Syntax</i> • H.263 303: - <i>Motion Compensation in H.263 Motion Compensation in H.263, Optional H.263 Coding Modes, H.263+ and H.263++</i> • ADPCM in Speech Coding • G.726 ADPCM • Vocoder:- Phase Insensitivity, Channel Vocoder, Formant Vocoder, Linear Predictive Coding, CELI, Hybrid Excitation Vocoder.	14
Multimedia Networks and QoS Support	• Basics of Computer and Multimedia Networks: <i>OSI Network layers, TCP/IP Protocols</i> • Multiplexing Technologies: <i>Basics of multiplexing, integrated services digital networks, synchronous optical network, asymmetric digital subscriber Line</i> • LAN and WAN: ◦ <i>Local area networks, wide area networks, asynchronous transfer mode, gigabit and 10-gigabit ethernets.</i> • Access Networks, • Common Peripheral Interfaces	10
Multimedia Wireless Networks, Heterogeneous Networks, and advanced QoS Support	• Quality of multimedia data transmission: <i>Quality of service, QoS for IP protocols, prioritized delivery</i> • Multimedia over IP: <i>Ip-Multicast, RTP (Real-time Transport Protocol), RTCP, Resource Reservation Protocol, Real time streaming protocol, Internet Telephony</i> • Wireless: <i>Analog Wireless networks, Digital wireless networks, TDMA and GSM, Spread spectrum and CDMA, analysis of CDMA, 3G Digital Wireless Networks, Wireless LAN</i>	10
Multimedia Applications	• Media-on-demand: <i>interactive TV (ITV) and set-top Box (STB), Broadcast</i>	

	<i>Schemes for Video-on-demand, Buffer Management</i>	
Topics in Multimedia Technologies	• Radio propagation Models: <i>Multipath Fading, Path loss</i> • How should we retrieve images efficiently? image searching systems and example projects • Multimedia security, digital rights management, etc.	
Total		60

References and materials

Textbook

1. Ze-Nian Li & Mark S. Drew. (2014). *Fundamentals of Multimedia*. Upper Saddle River, NJ: Pearson Education (*Main text for the course*)

Other resources for the course:

- Links to a variety of resources on the World Wide Web.
- An introduction to digital multimedia by Savage, T. M. and Vogel, K. E. 2008.
- Digital Multimedia by Nigel Chapman & Jenny Chapman. 2009.

References

1. Furht, B., Socek, D. and Eskicioglu, A.M., 2004. Fundamentals of multimedia encryption techniques. *Multimedia Security Handbook*, 4.
2. Guan, L., He, Y. and Kung, S.Y. eds., 2012. *Multimedia image and video processing*. CRC press.
3. Zeng, W., Yu, H. and Lin, C.Y. eds., 2011. *Multimedia security technologies for digital rights management*. Elsevier.

4.6 Year 3 Semester 2

Course Name	Business and Entrepreneurship
Code	CS3201
Type	Core
Contact Hours	48
Credit Units	4
Prerequisite	None
Level	Year 3 Semester 2

Course Description

This course provides students with an introduction to the potential and pitfalls of Entrepreneurship. Students will learn about the various methods for starting up, managing and financing a new venture. Students will get a chance to evaluate real world business plans and videos of entrepreneurs.

Course Objectives

At the end of this course student should be able to;

1. Develop knowledge and appreciation Entrepreneurship and Small Business Management through the study of new ventures, and management of small firms.
2. Explain the ethical and legitimacy challenges that face entrepreneurs with new ventures.

Learning Outcomes

At the end of this course student are able to:

1. Develop an understanding of the nature of entrepreneurship & determine whether you want to be an entrepreneur with your own business or a corporate entrepreneur (entrepreneur in someone else's business).
2. Understand how to identify opportunities (problems), develop creative solutions and build a viable business model around them.
3. Identify and understand the driving forces of new venture success.

Course Outline

Topic	LH	TH	PH
INTRODUCTION AND WELCOME!!! Introduce Case Study Teams Small Business: • Its Opportunities and Rewards • BRIE Model • Small Business Administration • Small Business Development Center • Franchise • Small Business Myths • 4 forms of Entrepreneurship	2	0	0
SMALL BUSINESS ENVIRONMENT:	2	0	0

Managing external relations • Focus on Small Business • BRIE Model • Build Legitimacy for your org • Social Networking • Basic Skills for handling crisis			
SMALL BUSINESS ENTREPRENEURS: Characteristics and competencies • Entrepreneur personalities • 5 operational competencies • Career paths • Family business • Entrepreneurial teams • Late career entrepreneurs	1	0	0
SMALL BUSINESS IDEAS: Creativity, opportunity and feasibility • Entrepreneurial opportunities • Evaluating viable ideas • Creativity methods of opportunity • 5 pitfalls that hinder innovation • Strategy Feasibility studies • Internet business	2	1	0
SMALL BUSINESS STRATEGIES – IMITATION WITH A TWIST; • Know decisions needed as a foundation for strategic planning • Forms of imitative and innovative businesses • Benefits that win over customers • How industry changes can affect strategies • SWOT analysis to identify strategic options	2	1	0
SMALL BUSINESS ENTRY: Paths to part-time entrepreneurship • Full time or Part time • Major paths to part time entrepreneurship • Finding success with BRIE model • How and when to delegate • Bootstrapping • Ethics of part-time entrepreneurship • Part-time to Full-time	1	1	0
SMALL BUSINESS ENTRY: Paths to full time entrepreneurship • 5 ways to get into small business management	2	0	0

• Rewards v pitfalls of starting a new business • Rewards v pitfalls of purchasing existing business • 4 methods of purchasing an existing business • Advantage v disadvantage of buying franchise • Inheriting a family owned business • Hired managers become owners of small business			
BUSINESS PLANS – SEEING AUDIENCES AND YOUR BUSINESS CLEARLY • Why and When to develop a business plan • Know how to tell the business plan story • Major sections of the classical business plan • Focus sections to meet specific needs • Identify major pitfalls when creating business plan • Master presenting business plan to others	2	1	0
MARKETING PLAN – SAYING HOW YOU WILL GET SALES • Importance of a Marketing Plan • Major methods for marketing research • Sales forecasting methods • Create a products differential advantage • Critical components of a marketing plan	2	0	0
SMALL BUSINESS PROMOTION: Capturing the eyes of your market; • Asses value proposition • Segment and further define target audience • Different approaches and methods to craft and convey promotional audience • Key skills involved in personal selling, especially closing the sale • Major approaches to customer relationship management • Develop a press relations program • Develop a public relations program	2	1	0
PRODUCT AND PRICING STRATEGIES • Characteristics of goods and services • Define the total product • Stages of new product development • Product life cycle • Why pricing is an important but difficult task for small business • Price elasticities, price and psychology, and other influencers and their impact on pricing • Different pricing strategies	2	1	0

ACCOUNTING - PROJECTING AND EVALUATING PERFORMANCE • Basic concepts of accounting • Requirements for small business accounting system • Content and format of common financial statements • Accounting information as a tool for managing your business effectively • Develop a complete set of craft budgets for your business • Using accounting information to make better business decisions	2	1	3
CASH – LIFEblood OF THE BUSINESS • Importance of managing your business money • Understand concept of money, cash and cash equivalents • Basics of managing cash flow • Reconcile bank and company book balances • Develop cash budget • Strategies for preventing and coping with cash flow problems • Strategies for coping with cash shortages	2	0	0
SMALL BUSINESS FINANCING – EQUITY, DEBT AND GIFTS • 3 types of capital financing – costs and trade-offs • Characteristics of a business that determines its ability to raise capital • Best type of financing for your business • Different needs for financial management at each stage of business life	2	1	3
ASSETS: INVENTORY AND OPERATIONS MGMT • Techniques to manage short-term assets • Calculate value of assets of your business • Technique for managing fixed assets • Advantages of renting or leasing capital equip • Techniques to manage and improve operations of your business	2	1	3
HUMAN RESOURCES • Hire part-time or full time • How to recruit good employees on your budget • Employee training • Employee needs and expectations • Fair compensation and benefits • How to handle family employees	2	1	0

SMALL BUSINESS PROTECTION – RISKS MANAGEMENT AND INSURANCE • Understand meaning and nature of business risk • Specific types of risks associated with different aspects of business operations • Techniques to manage risks to stay in level of risk tolerance • How insurance can be used to manage business risk • Techniques for sharing risk with other businesses and organizations	2	0	0
LEGAL ISSUES • Knowing when you need legal info and how to get it • Understand business structures of small business • Master the process of negotiation • Legal liabilities for your business • Contracts, terms and when you need one • Intellectual property	2	1	3
ACHIEVING SUCCESS IN THE SMALL BUSINESS • Stages of the small business life cycle • 4 types of firm-level growth strategies • Options – harvesting or closing small business • Critical Success factors of small business	2	1	0
Total hours	36	12	0
	60		

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.
- Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

Assessment Mark

Course works

- Exercises 40%
- Quizzes
- Tests

Final exams 60%

Basic Reading List

1. Spinelli, S. & Adams, R. (2012). The Entrepreneurial Mind: Crafting a Personal Entrepreneurial Strategy. New Venture Creation: entrepreneurship for the 21st century (9th Ed., pp. 35-48). New York: McGraw Hill.
2. Spinelli, S. & Adams, R. (2012). The Entrepreneurial Process. New Venture Creation: entrepreneurship for the 21st century (9th Ed., pp. 87-103). New York: McGraw Hill.

Course Name	Computer Science Project
Code	CS3202

Level	Year 3 Semester 2
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Description

A project devoted to an independent investigation and report on a computer science topic. The report should reflect the capacity of the student to apply theoretical and practical knowledge in information Technology.

A final year project is conducted under the supervision of member of the academic staff and an original copy shall be handed in by the student. Each student will be expected to orally present the findings, analysis, design and implementation of a system or application.

This course is research driven, involves a student choosing an appropriate area of specialization in the computer science in which the student feels comfortable.

Course Objectives

At the end of this course student should be able to:

1. Developing of a concept for a feasible software project development.
2. Defining requirements and design specifications.
3. Explain appropriate software engineering processes, tools and methods.
4. Professional software construction, testing, documentation.

Learning Outcomes

The main learning outcomes of this course are:

- a) Students should concretize skills of software development learnt in previous courses.
- b) Students should realization of skills in analysis, design and reporting of software processes.
- c) Students should demonstrate skills of project documentation.

Course Content

- a) Concept of report writing
- b) Concept of literature reviews
- c) Concept of problem analysis and design
- d) Concept of development of software based on analysis.
- e) Making conclusion on done work

Mode of Delivery

- a) Student presentations
- b) Continuous supervision by staff members.

Method of Assessment

The student's project will be assessed out of 100%, which will be distributed as follows:

- a) Project Proposal - 20%
- b) Project report – 50%
- c) Project Presentation – 30%

Key Reading List

No reading list though several references will have to be used by the help of the supervisor.

Course Name	Compiler Construction
Code	CS2208

Level

Year 3 Semester 2

Course Description

Introduces the basic techniques used in translating programming languages: scanning, parsing, lowering, code selection, and register allocation. Students build a complete compiler for successively richer languages. The very first is a tiny language of arithmetic statements, and by the time the course is complete the language includes objects, inheritance, and first-class functions.

Course Objectives

1. Explain how to build a compiler for a (simplified) (programming) language
2. Demonstrate how to use compiler construction tools, such as generators for scanners and parsers
3. Explain virtual machines, such as the JVM and Java bytecode
4. Write LL(1), LR(1), and LALR(1) grammars (for new languages)
5. Be familiar with compiler analysis and optimization techniques

Learning Outcomes/ Goals

- Understand the structure of compilers
- Understand the basic techniques used in compiler construction such as lexical analysis, top-down, bottom-up parsing, context-sensitive analysis, and intermediate code generation
- Understand the basic data structures used in compiler construction such as abstract syntax trees, symbol tables, three-address code, and stack machines

Course Outline

Topic	LH	TH	PH
Introduction to Compilers: • The role of language translation in the programming process; • Comparison of interpreters and compilers, language translation phases, • machine-dependent and machine-independent aspects of translation, • language translation as a software engineering activity	2	0	2
Lexical Analysis: • Application of regular expressions in lexical scanners,	3	0	2

• hand coded scanner vs. automatically generated scanners, • formal definition of tokens, • implementation of finite state automata.			
Syntax Analysis: • Revision of formal definition of grammars, • BNF and EBNF; • bottom-up vs. top-down parsing, • tabular vs. recursive-descent parsers, • error handling.	3	0	2
Parsers Implementation: • automatic generation of tabular parsers, • symbol table management, • the use of tools in support of the translation process.	2	0	2
Semantic Analysis: • Data type as set of values with set of operations, • data types, • type- checking models, • semantic models of user-defined types, • parametric polymorphism, • subtype polymorphism, • type-checking algorithms.	3	3	3
Intermediate Representation, code generation: • Intermediate and object code, • intermediate representations, • implementation of code generators.	2	3	4
Code optimization: • Machine-independent optimization; • data-flow analysis; • loop optimizations; • machine-dependent optimization	3	3	3
Error Detection and Recovery,	2	0	2
Error Repair, Compiler Implementation	2	0	2
Compiler design options and examples: C Compilers	2	3	2
	24	0	24
Total hours	60		

Mode of Delivery

Utilize a question/answer methodology (Socratic method of teaching) in which Lecture, Slides, Handouts, case studies, current news articles and assignments are used to initiate discussion in which students themselves do problem solving to arrive at conclusions about topics.

Blackboard is used to broadcast information and class updates. Use break out group sessions as well as to build communication and team working skills as well as student presentations to develop individual communication skills.

Mode of Assessment

Assessment	mark
Course works	
• Exercises	40%
• Quizzes	
• Tests	
Final exams	
	60%

Basic Reading List

1. Compilers Principles, Techniques and Tools Author(s)/Editor(s): Alfred V. Aho, Ravi Sethi and Jeffrey D. Ullman, Publisher: Addison Wesley Longman, 1986.
2. W. Appel, Modern Compiler Implementation in Java, Prentice Hall, 2002.
3. D. Watt, Brown, Programming Language Processors in Java: Compilers and Interpreters, Prentice hall, 2000

Course Name	Parallel Processing
Code	CS3205

Level	Year 3 Semester 2
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Course Description

Despite the extraordinary advances in computing technology, we continue to need ever greater computing power to address important fundamental scientific questions. Because individual compute processors have essentially reached their performance limits, the need

for greater computing power can only be met through the use of parallel computers. This course is intended for students who are interested in learning how to take advantage of parallel and distributed computing with the focus of writing parallel code for processor intensive applications to be run on clusters, the grid, the cloud, or shared infrastructure such as that provided by Compute Canada. The objectives of this course are to give the students an understanding of how they can use parallel computing in their research and enable them to write parallel code for their high-performance computing applications. Extensive use of pertinent and practical examples from scientific computing will be made throughout. The programming languages used will be Matlab, Maple, sage, python, Fortran, or C. Both the shared and distributed paradigms of parallel computing will be covered via the OpenMP and MPI libraries.

Course Objectives

At the end of this course student should be able to;

- a) Implement small applications on shared-memory multiprocessor (Linux SMP) using pthreads.
- b) Understand concepts of parallel algorithms
- c) Explain elementary topologies and communication techniques
- d) Understanding of compiler concurrentization concepts

Learning outcome

This course introduces critical methods and techniques related to parallel computing. Particularly, the course focuses on hardware, algorithm, and programming of parallel systems, providing students a complete picture to understand pervasive parallel computing.

Through this course, the students will have a deep understanding of how parallel systems are designed and what are the fundamental methods to program and analyse them. By the end of this course, students will be able to:

- a) Explain how large-scale parallel systems are architected and how massive parallelism are implemented in accelerator architectures;
- b) Write parallel programs for large-scale parallel systems, shared address space platforms, and heterogeneous platforms;
- c) Design efficient parallel algorithms and applications;
- d) Be conversant with performance analyse and modelling of parallel programs;

Relation to Program Learning Outcomes

This course relates to the following program learning outcomes:

- a) An ability to apply knowledge of computing and mathematics appropriate to the discipline;
- b) An ability to analyse a problem and identify the computing requirements appropriate for its solution; an ability to design, implement and evaluate a computer-based system, process, component or program to meet desired needs;
- c) An ability to apply mathematical foundations, algorithmic principles and computer science theory to the modelling and design of computer based systems in a way that demonstrates comprehension of the trade-offs involved in design choices;
- d) An ability to apply design and development principles in the construction of software systems of varying complexity.
- e) An ability to function effectively as a member of a team in order to accomplish a common goal;

Course Outline

1. Computer Architecture and Systems
 - a. Memory Subsystem & Cache
 - b. Shared Memory Systems: Memory Consistency Models and Cache Coherence
 - c. Distributed Memory Systems: Message Passing and Network Topologies
 - d. Synchronization
 - i. Primitives and Implementation: Locking, Barrier, Semaphore
 - ii. Probs: Deadlock, Livelock, Starvation, Priority Inv., Race Cond.
 - e. Architecture: Multicore, Manycore, GPU and Vector Machines
 - f. Multithreading
 - g. Exascale challenges: Fault tolerance, Communication (memory hierarchy) and Power management
2. Data/Task Distribution
 - a. Load Balancing: Static and Dynamic
 - b. Graph Partitioning and Algorithms
 - c. Locality in Simulation
 - i. cyclic, blocked, block-cyclic data distribution
 - ii. surface to volume minimization
3. Languages/Libraries
 - a. Data Parallel Languages: CUDA
 - b. Global Address Space Languages: UPC
 - c. Compiler-aided Parallelization: OpenMP
 - d. Message-Passing Library: MPI
4. Performance modeling of Networks: PRAM, Alpha/Beta Model, LogP/LogGP, Roofline Model
5. Applications/Algorithms
 - a. Dense/Sparse Linear Algebra
 - i. Algorithms:

1. Matrix Multiplication (Canon's algorithm, SUMMA, Vasily's study)
2. Gaussian Elimination, LU
3. Sparse GE/LU, sparse matrix storage
- ii. Autotuning
- b. N-Body/Particle Simulations: Barnes Hut and FMM
- c. Finite Difference Approximations to PDEs
 - i. Jacobi, SOR, conjugate gradients, Multigrid
 - ii. FFT
- d. Structured and Unstructured Grids: AMR
- e. Graph Algorithms: scans on GPU, pointer-jumping, list ranking and parallel prefix

Table of chapters

Topic	LH	TH	PH
Computer Architecture and Systems	4	2	4
Data/Task Distribution	4	2	4
Languages/Libraries	4	2	4
Performance modelling of Networks	4	2	4
Applications/Algorithms	8	4	8
Total hours	24	12	24

Mode of Assessment

Tests and Quizzes	5%
Assignments	15%
Course Project	20%
Final Exam	60%
Total	100%

Reading List

Main Texts

1. Lin & Snyder, 2009. Principles of Parallel Programming
2. Parallel Programming in C with MPI and OpenMP by M J Quinn

Supplemental

1. Storti & Yurtoglu, 2016. CUDA for Engineers: An Introduction to High-Performance Parallel Computing

2. Jordan & Alaghband, 2003. Fundamentals of Parallel Processing
3. Introduction to Parallel Computing by Ananth Grama, George Karypis, Vipin Kumar, and Anshul Gupta.
4. Programming Massively Parallel Processors by D. Kirk and W. Hwu

References

1. Kaushik Datta, Mark Murphy, Vasily Volkov et. al., Stencil Computation Optimization and Auto-tuning on State-of-the-Art Multicore Architectures, Proc. SC2008, Nov. 2008
2. S. Williams, A. Waterman, D. Patterson, Roofline: An Insightful Visual Performance Model for Floating Point Programs and Multicore Architectures, Communications of the ACM (CACM), April 2009
3. John L. Hennessy and David A. Patterson, Computer Architecture~ A quantitative Approach (Fourth Edition), Morgan Kaufman Publishers, San Francisco, CA, 2007

Course Name	Design Principles of HCI
Code	CS3209

Level	Year 3 Semester 2
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Course Description

This course is designed to give a foundational understanding of how people interact with computers and computing technology, and will provides a set of basic skills for evaluating and designing for this type of interaction.

The study of human-computer interaction enables system architects to design useful, efficient, and enjoyable computer interfaces. This course teaches the theory, design procedure, and programming practices behind effective human interaction with computers, smart phones and tablets.

There is examining of interaction design, implementation, and evaluation. The design process requires a solid understanding of the theory behind successful human-computer interaction, as well as an awareness of established procedures for good user interface design, including the 'usability engineering' process. Iterative evaluation is an important aspect of this procedure, and we will learn and practice prototyping and evaluation using scenario-based case studies. Students will apply their gained knowledge in a series of practical assignments using the Android ecosystem that highlight selected portions of the

design cycle, as well as familiarize them with sound programming practices and effective tools and techniques to create successful user interfaces.

Course Objectives

At the end of this course student should be able to;

- a) Understand Human Computer Interaction (HCI) principles,
- b) Understand UI problems, issues, and challenges
- c) Identify the impact of system design on users' attitudes and behaviours,
- d) Apply user-centered design processes in designing and building systems.
- e) Become familiar with important HCI research issues and technology trends

Learning Outcomes/ Goals

At the end of this course student are able to;

- a) Critically discuss common methods in the user-centered design process and the appropriateness of individual methods for a given problem.
- b) Use, adapt and extend classic design standards, guidelines, and patterns.
- c) Employ selected design methods and evaluation methods at a basic level of competence.
- d) Develop prototypes at varying levels of fidelity, from paper prototypes to functional, interactive prototypes.

Course Outline

Revision content: HTML 5, CSS 3, JavaScript/PHP/MySQL/AJAX/jQuery (web and mobile)

Topic	LH	TH	PH
What is Interaction Design? Understanding and conceptualizing interaction	2	2	2
Understanding users	2		2
The process of Interaction Design	2	2	2
Establishing requirements	2		2
Prototyping such as Paper Prototyping and Mock-ups, Video Prototyping, Creating and Comparing Alternatives	4	2	4
Evaluation: Heuristic Evaluation, Design Heuristic	4		4
Observing users	2	2	2

Visual Design and Information Design, Fonts/Maps/Graphics	4	2	4
Emerging Trends	2	2	2
Total hours	24	12	24

Mode of Delivery

- Formal lectures, interactive lectures, group tutorials.
- Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

Assessment mark

Course works

- Exercises 10%
- Quizzes and Tests 5%
- Project 20%

Final exams 60%

Basic Reading List

Required

1. Shneiderman et al, 2017. Designing the User Interface: Strategies for Effective Human-Computer Interaction, 6th Edition

Supplemental

1. Rodgers, Y., Sharp, H., Preece, J., 2011. Interaction Design: Beyond Human-Computer Interaction, 3rd ed. Wiley.
2. Moggridge, B., 2007. Designing Interactions. Cambridge, MA: The M.I.T. Press.
3. Dix, Finlay, Abowd & Beale, 2004. Human-Computer Interaction, 3rd Edition

Course Name Data Science
Code CS3208
Type Elective

Course Description

Data Science is the study of the generalizable extraction of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases and other branches of computer science along with a good understanding of the craft of problem formulation to engineer effective solutions. This course will introduce students to this rapidly growing field and equip them with some of its basic principles and tools as well as its general mind-set.

Students will learn concepts, techniques and tools they need to deal with various facets of data science practice, including data collection and integration, exploratory data analysis, predictive modelling, descriptive modelling, data product creation, evaluation, and effective communication.

The focus in the treatment of these topics will be on breadth, rather than depth, and emphasis will be placed on integration and synthesis of concepts and their application to solving problems. To make the learning contextual, real datasets from a variety of disciplines will be used.

Course Objectives

This course addresses the growing need from both industry and academia (e.g., medical and bio informatics, financial, law enforcement, economics, decision support, social networks) for big data analytic skills including, data management, data mining, machine learning and data visualization. The aim is to bring students, with basic programming and data structure background, to be abreast with common tools used for Data Science application development. This course serves as an introduction to the basics of data science including programming for data analytics, file management, relational databases, classification, clustering and regression. The foundation is laid for big data applications ranging from social networks to medical and business informatics.

Learning Outcomes

At the conclusion of the course, students should be able to:

- a) Describe what Data Science is and the skill sets needed to be a data scientist.
- b) Explain in basic terms what Statistical Inference means. Identify probability distributions
- c) Explain the significance of exploratory data analysis (EDA) in data science. Apply basic tools (plots, graphs, summary statistics) to carry out EDA.
- d) Apply basic machine learning algorithms (Linear Regression, k-Nearest Neighbours (k-NN), k-means, Naive Bayes) for predictive modelling. Explain why Linear

Regression and k-NN are poor choices for Filtering Spam. Explain why Naive Bayes is a better alternative.

- e) Identify and explain fundamental mathematical and algorithmic ingredients that constitute a Recommendation Engine (dimensionality reduction, singular value decomposition, principal component analysis). Build their own recommendation system using existing components.

Course Outline

Case studies are to be included in all topics.

Topic	LH	TH	PH
Introduction: What is Data Science? – Intro to Big Data and Data Science – Current landscape of perspectives – Skill sets needed	2		2
Statistical Inference – Populations and samples – Statistical modelling, probability distributions, fitting a model – Intro to R or MATLAB	4	1	8
Exploratory Data Analysis and the Data Science Process – Basic tools (plots, graphs and summary statistics) of EDA – Philosophy of EDA – The Data Science Process – Three Basic Machine Learning Algorithms – Linear Regression – k-Nearest Neighbours (k-NN) – k-means	4	1	2
Machine Learning Algorithm and Usage in Applications – Motivating application: Filtering Spam – Why Linear Regression and k-NN are poor choices for Filtering Spam – Naive Bayes and why it works for Filtering Spam – Data Wrangling: APIs and other tools for scrapping the Web	4	1	2
Feature Generation and Feature Selection (Extracting Meaning from Data) – Motivating application: user (customer) retention – Feature Generation (brainstorming, role of domain expertise, and place for imagination) – Feature Selection algorithms	4		4

Recommendation Systems: Building a User-Facing Data Product – Algorithmic ingredients of a Recommendation Engine – Dimensionality Reduction – Singular Value Decomposition – Principal Component Analysis – build your recommendation system	4	1	2
Mining Social-Network Graphs – Social networks as graphs – Clustering of graphs – Direct discovery of communities in graphs – Partitioning of graphs – Neighbourhood properties in graphs	4	1	2
Data Visualization – Basic principles, ideas and tools for data visualization – Examples of inspiring (industry) projects	2	1	2
Data Science and Ethical Issues – Discussions on privacy, security, ethics	2		
Total	30	6	24

Mode of Delivery

1. Formal lectures, interactive lectures, group tutorials.
2. Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

The final grade will be determined based on performance on each of the following items; the percentages in parenthesis show the weight each item carries to the final grade.

- Test and quizzes (5%)
- Assignments (15%)
- Project (20%)
- Final examination (60%)

Basic Reading List

The following book will be used as a textbook and primary resource to guide the discussions, but will be heavily supplemented with lecture notes and reading assignments from other sources.

1. Cathy O'Neil and Rachel Schutt. Doing Data Science, Straight Talk from the Frontline. O'Reilly. 2014.

Additional references and books related to the course:

1. Jure Leskovek, Anand Rajaraman and Jeffrey Ullman. Mining of Massive Datasets. v2.1, Cambridge University Press. 2014. (free online)
2. Kevin P. Murphy. Machine Learning: A Probabilistic Perspective. ISBN 0262018020. 2013.
3. Foster Provost and Tom Fawcett. Data Science for Business: What You Need to Know about
4. Data Mining and Data-analytic Thinking. ISBN 1449361323. 2013.
5. Trevor Hastie, Robert Tibshirani and Jerome Friedman. Elements of Statistical Learning, Second Edition. ISBN 0387952845. 2009. (free online)
6. Avrim Blum, John Hopcroft and Ravindran Kannan. Foundations of Data Science.
7. Mohammed J. Zaki and Wagner Miera Jr. Data Mining and Analysis: Fundamental Concepts and Algorithms. Cambridge University Press. 2014.
8. Jiawei Han, Micheline Kamber and Jian Pei. Data Mining: Concepts and Techniques, Third Edition. ISBN 0123814790. 2011.

Course Name	UNIX Shell Programming
Code	CS3206
Type	Elective

Level	Year 3 Semester 2
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Course Description

This course is designed as a beginners guide to the Unix operating systems which have long been popular in academic contexts. The course covers the most basic and frequently used Unix / Linux commands and introduces the proper use of UNIX commands, maintain files, manage processes, and code UNIX shell scripts. Most of the built-in shell commands are introduced together with the main program control structures.

Course objectives

At the end of the course, students should be able to:

- a) Use the UNIX operating system to develop for information handling and software development;
- b) Develop engineering applications using c language in a UNIX environment.

Course Outcomes

At the end of the course, students shall be able to;

1. Discuss the architecture, networking and basic commands of UNIX. (Understand)
2. Implement various file processing commands used in UNIX. (Apply)
3. Apply Regular expression to perform pattern matching using utilities like grep,sed and awk. (Apply)
4. Construct various shell scripts for simple applications. (Apply)
5. Explain the process management using system calls UNIX environment .(Understand)

Course Outline

Chapter 1. (6 hours)
Introduction to Unix:- Architecture of Unix, Features of Unix , Basic Unix Commands – Unix Utilities:- Introduction to unix file system, vi editor, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands – Text processing utilities and backup
Chapter Two (11 hours)
Introduction to Shells:-Unix Session, Standard Streams, Redirection, Pipes, tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization. Regular expressions, Filters and Pipes, Concatenating

files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count characters, words or lines, Comparing Files.

Chapter Three (10 hours)

grep:-Operation, grep Family, Searching for File Content.

sed:-Scripts, Operation, Addresses, commands, Applications, grep and sed.

awk:-Execution, Fields and Records, Scripts, Operations, Patterns, Actions,

Associative Arrays, String Functions, Mathematical Functions, User Defined Functions, Using System commands in awk, Applications of awk, grep and sed

Chapter Four (10 hours)

Interactive Shells - Korn Shell, C Shell and BASH - Shell Features, Special Files, Variables, Output, Input, Exit Status of a Command, eval Command, Environmental Variables, Options, Startup Scripts, Command History, Command Execution Process. Shell Programming - Korn Shell, C Shell and BASH

Basic Script concepts, Expressions, Decisions: Making Selections, Repetition, special Parameters and Variables, changing Positional Parameters, Argument Validation, Debugging Scripts, Script Examples.

Chapter Five (8 hours)

Process management:- Creation, Hierarchies, Sending signals to processes, exec, termination, Zombie, waitpid etc - Network management:- tools, Client server mechanism, address resolution, ping, telnet, ftp, dns and squid – X Window System:- Overview, Architecture, starting and stopping X, X clients and display

Mode of Delivery

3. Formal lectures, interactive lectures, group tutorials.
4. Interactive, communicative, discussion based learning, project based learning and task based learning.

Mode of Assessment

The final grade will be determined based on performance on each of the following items; the percentages in parenthesis show the weight each item carries to the final grade.

- Test and quizzes (5%)
- Assignments (15%)
- Project (20%)
- Final examination (60%)

Basic Reading List

The following book will be used as a textbook and primary resource to guide the discussions, but will be heavily supplemented with lecture notes and reading assignments from other sources.

1. Sumitabha Das , “Unix the ultimate guide”, TMH. 2nd Edition. .
2. Behrouz A. Forouzan, Richard F. Gilberg,” Unix and shell Programming.”, Cengage Learning
3. Kernighan and Pike, “Unix programming environment”, PHI. / Pearson Education
4. Graham Glass, King Ables,” Unix for programmers and users”, 3rd edition, Pearson Education
5. Maurice J. Bach, “The Design of the Unix Operating System”, First Edition, Pearson Education, 1999

4.1 Quality of the Academic Staff

The Faculty encourages inter-disciplinary teaching across other faculties and visiting lectureship between universities (with which Mbarara University of Science and Technology has established such cordial relationships), as well as amicable relationship between staff and its students so as to foster confidence among the students: which confidence will be of potential application when they leave the University and go out in the world to be on their own. Table 4.1 describes the academic staff responsible for the programme.

Table 4.1: Academic staff

No	Names	Designation	Qualifications	Awarding Institution	Specialisation	Year of completion
1	Richard Ssembatya	Senior Lecturer	BSc MSc PhD	MUST MUK UCT	HCI and ICT for Development	2001 2006 2014
2	Evarist Nabaasa	Senior Lecturer	BSc MEng. Comp. PhD	MUK Hunan Univ.China MUST	Embedded Systems and Web Technologies	2000 2006 2015
3	Angella Musiimenta	Senior Lecturer	BSc MSc PhD	MUST Univ. of Leeds Univ. of Manchester	Health Informatics	2003 2005 2011
4	John Businge	Senior Lecturer	BSc MSc PhD Post doc	MUK Croningen Eindhoven Univ. of Techn.	Software Engineering	2002 2006 2013
5	Fred Kaggwa	Lecturer	BSc. MSc. PhD.	MUST Twente. University Netherlands MUST	System and Network Security and	2007 2011 2016
6	Simon Kawuma	Lecturer	BSc. MSc.	MUST Univ. of Southampton, UK Norwegian Univ. of Sc & Tech	Software Engineering and Embedded Systems	2009 2013

			PhD.			2019
7	Pius Ariho	Lecturer	BSc. PhD	MUST University of New Brunswick, America	Computational Mathematics	2007
8	Jane Katusiime	Lecturer	BSc. MSc PhD completion). (in	MUST Malmö University, Sweden Humboldt University, Berlin	HCI	2007 2013 2019
9	Kenneth Baguma	Lecturer	BSc. MSc.	UCU, Mukono Uganda Martyrs, Nkozi	Information Systems and technologies	2009 2014
10	Robert Mugonza	Lecturer	BSc. MSc.	MUST Leeds	Robotics and Embedded system	2008 2013
11	Ruth Nakato	Lecturer	BSc. MSc.	MUK MUK	Information Systems	2004 2008
12	Martin Ngobye	Lecturer	BSc. Phy/Maths MSc. Eng.	MUK Hunan, China	Digital Electronics	2002 2005
13	Aaron Atuhe	Lecturer	BSc. MSc.	VCU KIU	Data mining	2007 2012
14	David Bamutura	Lecturer	BSc. MSc.	UMU UoK	Data Science	2008 2010
15	Walter Akello	Lecturer	BSc. MSc.	Kharkou Nat. Univ. Of Radio Electronics	Data communication and Networks	2004 2006
16	Aggrey Obbo	Lecturer	BSc. Mathematics & Statistics BSc. MSc.	MUK MUK	Networks and Computational Mathematics	2001 2007

17	Deborah Natumanya	Lecturer	BSc. MSc.	MUST MUST	Information Systems and Education Technologies	2014 2016
18	Richard Ntwari	Assistant Lecturer	BSc. MSc.	MUST UMU, Nkozi,	System security and Management Science	2010 2017
19	Tezira Wanyana	Assistant Lecturer	BSc. MSc.	MUST University of Madras, India	AI and Big Data	2010 2016
20	Tonny Engwau	Assistant Lecturer	BSc. MSc.	MUST MUK	GIS	2011 2019
21	Yonasi Safari	Assistant Lecturer	BSc. CS PGD Medical Informatics	MUK Kwazulu, S. Africa Kwazulu, S. Africa	Parallel/signal processing	2011 2015 2016
22	Mwavu Rogers	Assistant Lecturer	BSc. MSc.	MUST MUST	Big Data and IoT	2014 2016
23	Francis Kamuganga	Teaching Assistant	BSc. MSc.	MUST MUK	AI and Computer vision	2010 2017
24	Richard Kimera	Teaching Assistant	BSc. IT MSc. MSc.	MUST Hardong Global Univ., South Korea	Mobile Technologies	2011 2013 2016
25	Josephine Ayebare	Teaching Assistant	BSc. MSc.	UCU, Mukono Stockholm Univ., Sweden	Wireless Technologies	2007 2017
26	Moreen Kabarungi	Teaching Assistant	BSc. MSc.	MUST MUST	Information Systems and Technologies	2010 2016
27	Ambrose Izaara	Teaching Assistant	BSc. MSc.	MUST MUST	Networks and Web technologies	2009 2016
28	Gloria Munguci	Teaching Assistant	BSc. MSc.	UCU, Mukono MUST	Health Informatics	2018

5.2 Quality of the support staff

The Faculty has two administrators one with a Bachelor's degree in Administrative and Secretarial Science and Masters in Management Studies, and the other secretarial studies. The faculty also has two ICT administrators who are pursuing their Master of Science in

Information systems and Security. In addition, the Faculty also enjoys the general services provided by the university such as ICT support from the ICT Department, cleaning, library, and any other service the Faculty may require from the Estates Department.

5.3 Student advice/support

All students admitted Mbarara University of Science and Technology undergo one-week orientation organised by the Office of the Dean of Students and students' leaders, to help the freshers get into the system of the University. During this time, they meet various University officials who address and counsel them on the new type of life and study they are going to face. Among the University officials they meet the Library staff who takes them through services offered by the Library and how to access e-resources.

During the faculty orientation, students are briefed of the faculty regulations and the expectations they are expected to meet. They are given a Faculty Handbook which contains the Faculty program and regulations and thereafter, they are ready to start their lectures.

5.3.1 Feedback from Students Enrolled

In the current set up, each class has 2 student representatives. These representatives are in constant contact with the Lecturers, Head of Department and the Dean, in case there are any quality related matters in a particular class, the representatives must report to responsible authorities.

At the end of the semester, all students are given evaluation forms to assess the lecturers. The form is in a form of a questionnaire, which helps the students to respond to several quality related matters like staff punctuality, delivery mode, course content and the general perceived usefulness of the course unit.

5.3.2 Class Meetings

The head of department holds meetings with every class at least once every semester. In this meeting, general quality issues are addressed. Students are also given a chance to raise any questions that are answered and/or addressed by the authorities.

5.3.3 Industrial Training Placement Reports

Students will be exposed to the industrial set up via industrial training. Reports from the placements will be used to gauge the effectiveness of the curriculum. They are also to be used to identify and address the deficiencies in the curriculum.

5.3.4 Tracer Studies

Mbarara University of Science and Technology through the Quality Assurance Directorate (QAD) devised a way of keeping in contact with its alumni together with their employers. This is one way of making a tracer study of its graduates. The Department of Computer

Science like other departments at MUST also use outputs of the tracer studies to gauge the quality of the program and whenever necessary, improve it.

5.4 Facilities and Infrastructures

The Faculty of Computing and Informatics (FCI) have enough resources and sufficient infrastructure to sufficiently run this revised programme. The space in the faculty comprises of three computer laboratories, five staff offices and one software incubation centre.

5.4.1 Computer Laboratories

The University has six general laboratories (strictly for students practice), teaching laboratories and specialized laboratories. These laboratories are shared among all the faculties and are scheduled by the Administrators according to the computer courses offered. In all, the faculty has two other specialised laboratories. Currently, the computer students' ratio of 1:3 is observed which is adequate for the practical components of the curriculum.

5.4.2 ICT Support Staff

No	Names	Designation	Qualifications	Awarding Institution	Specialisation
1	Amos Baryashaba	Head, Computing Services	BSc. Computer Science MBA Msc. HIT	MUST MUST MUST	Systems and Networks
2	Lasto Mubiru	Systems Administrator/ Network Administrator	BSc. Computer Science	MUST	System Security
3	Owen Muhangi	Systems Administrator/ WebMaster	BSc. Computer Science	MUST	Web Technologies
4	Peter Bamba	Computer Technician	Bachelor of Information Technology Diploma in Computer Science & IT	UMU Uganda Technical College, Bushenyi	Hard ware repair and maintain ace

5	Kelly Tumwine	Senior Computer Technician	Bachelor of Information Technology	BSU	Hard ware repair and maintain ace
6	Martin Kijumi	Computer/Lab Technician	Diploma in Information Technology	UMU	Systems Administration

5.4.2 Software

On top of the physical computers, students need software for the different practical sessions. Computers in specialised laboratories are installed with different software depending on their focus. Most of the software is available as free distributions for academic purposes. The faculty and department therefore have (and can access) enough Open Source Software (OSS) that can run the practical aspects of the program.

5.5.3 Library Services

Mbarara University of Science and Technology Main Library is well stocked with enough books for literature review; it also provides access to e-books, e-journals, and print journals, a well-stocked reference section and connections to many remote e-databases. The Online Journals which are also accessed through the main University Library provides a range of products from abstracts to full text papers. The University Library has also acquired a wide range of e-recourses which are to be accessed even from outside the University and this has promoting e-learning.

5.5.3.1 Library Support Staff

No	Names	Designation	Qualifications	Awarding Institution	Specialization	Year of completion
1	Wilson Adriko	Ag. Librarian	MSc.IS	MUST	Open-source software adoption and ICT4D	2017
2	Lawrence Kamihanda	Librarian I	MSc. Info.Sci.	MUK	Library Infrastructure	2000
3	Nabada Agnes	Librarian II	BLIS	MUK	Library User information needs	2007
4	Kansiime Hope	Assistant Librarian	MSc. Info.Sci.	UCU Mukono	information/Records management	2017
5	Stephen Lwanga	Assistant Librarian	BIT	MUST	IT Application in Libraries	2009
6	David Waswa	Assistant Librarian	SWASA-PGD in HR	UMI	Human Resource mgt	2012

7	Nabukenya Judith	Assistant Librarian	BLIS	Kabale Univ.	Evaluation of library services and resources	2019
8	Unity Atukunda	Assistant Librarian	BLIS	Kabale Uni	Electronic Resources Utilization	2014
9	Bernard Kabonekye	Library Assistant	Master of Arts	MUST	Community Developmet	2017
10	Sylvia Keitesi	Assistant Librarian	DLIS	Kabale Univ	Library Systems	2013
11	Judith Kobusingye	Assistant Librarian	DLIS	Kabale Univ	Library Systems	2005
12	Zaccheas Tigahwa	Library Assistant	CLIS	MUK	Library Systems	2003
13	Edson Ben Kikundi	Library Assistant	DRAM	MUK	Library Systems	2013
14	Anita Asasira	Data Entry Clerk	MSc. IS	UMU Nkozi	ICT4D	2014
15	Haawa Kyohairwe	Data Entry Clerk	BIT	MUST	Application of IT in Business	2013

5.5.4 Tuition Fees

Privately sponsored students shall pay all the required fees to the University as determined by the senate and approved by council periodically.

1. Bachelor of Science in Computer Science (Course duration is 3 years)

- a) Tuition fees Shs. 2,000,000= p.a.
- b) Functional fees Shs. 520,000= p.a.

Other information regarding fees structure, and for international students, please visit the website: <https://www.must.ac.ug/2019-2020-fees-structure/>

6.0 Quality Assurance

6.1 Students evaluation

At the end of the semester, all students are given evaluation forms to assess the lecturers. The form is questionnaire which helps the students to respond to several quality related matters like Lecturer punctuality, course content, course delivery method, the general perceived usefulness of the course unit and also point out other areas that they wish the Faculty should look into and make some necessary changes.

Once a lecturer is evaluated poorly, the Dean and the head of department will call him/her and they discuss the comments and if it is very serious the person is warned in writing. If the lecturer does not improve once s/he is warned, there is a possibility of not renewing the contract and/or termination of employment.

6.2 Conclusive opinion

The curriculum is designed in such a way that the implementation of the program structure shows in Chapter 2 – 2.2 Program Content and 2.3 Program organisation takes both theory and practical oriented approaches.

In the new design methods of delivery tends to focus more on the student hence changing from the traditional teacher centred to student centred approach. All the teaching and learning processes, lecturers are encouraged to apply the flipped classroom method and maximally utilise the practical hours in order to apply the practical oriented approach. The lecturers will give students topics to research and students' group discussions and presentations are made during lectures, hence encouraging the student-centred approach. The curriculum is designed in a systematic manner that year one semester one gives a baseline for semester two, respectively. These course units are continuously assessed, a minimum of two assessment tests and two course works are administered within the semester period. At the end of every semester qualified students are assessed with a final exam, and also students evaluate the lecturer as mentioned above.